MEANING, OPACITY, AND FUNCTION: AN OBSERVER-RELATIVE CLASSIFICATION OF ARTIFACTS

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Abstract. Whether an artifact is understood, merely apprehended, or functional depends on the observer confronting it rather than on the artifact alone. We take three binary predicates over a domain of artifacts – structured meaning M, residual opacity C, and functionality K, each read relative to an agent – and study the finite datum they induce: which of the seven regions they carve are inhabited for that agent.

Two modelling choices are stipulated before anything is derived: the semantic predicates need not exclude one another, and denotation is confined to their overlap $\Theta = M \cap C$. The admissible data are then exactly the covers of the predicate set, so their number – 109 labelled, 34 up to relabelling – is the classical cover count of a three-element set and not a new enumeration.

Three results follow. The types with empty symbolic carrier are exactly those omitting the two overlap cells, seventeen in number; since a query's denoting token must lie in that carrier – a requirement that is definitional – an agent of such a type poses no query whatever architecture it carries. The six-coordinate structural profile that a type fixes is realizable if and only if it satisfies four explicit Boolean conditions, whence 39 of the 64 profiles occur. And architectures inducing the same compositional closure may still differ in the generation depth they assign its elements and in the pairs on which their composition is defined, two being the least size of the functional set at which this can happen. The predicates are supplied model data throughout: identifying the intended assignment from external evidence lies outside the present formalism.

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Key words and phrases: observer-relative classification; semantic opacity; functionality; region systems; agent modelling; structured ambiguity; denotation; compositional closure.

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1. Introduction

1.1. **Context.** Every working epistemic system – a research group, a software stack, a learned model together with its operators – maintains a domain of *artifacts*: data, symbols, concepts, functions, entire subsystems. Each artifact stands in a definite semantic relation to the system that handles it. Some artifacts are *understood*: their content is structured, their position among other artifacts is stable, their properties can be enumerated, and their behavior under known operations is predictable. Others are merely *registered*: the system knows they exist, can name them, perhaps even invoke them, yet has no access to their internal structure. And, independently of either condition, an artifact may or may not *do* something: it may act on other artifacts and produce outputs, or it may be inert.

The middle case – used, demonstrably working, yet not understood – is not marginal. Trained models invoked through interfaces, cryptographic routines called as black boxes, legacy functions executed daily although no one can reconstruct their rationale: such artifacts are functional without being transparent, a condition analyzed in the philosophy of computation as *epistemic opacity* [Hum09]. The condition has a further twist: it is observer-relative. The same encryption function is a transparent tool for the cryptographer who understands and executes it, and an opaque but perfectly usable operator for the end user who invokes it through an API. Any classification that treats semantic status as a property of the artifact alone erases this relativity.

1.2. **Tension.** Existing formal frameworks each capture a piece of this picture. Rough set theory [Paw82] classifies a domain, relative to available information, into a positively characterized part, an uncertain boundary, and an exterior – but it *derives* that classification from an indiscernibility relation over whatever attributes are supplied, and studies the derivation in general, where what follows stipulates three attributes with a fixed reading and studies the cell structure they determine. The difference is in what is taken as given and not in expressive range: supplied with the right attributes, rough set theory represents the same distinctions, as Section 9.1 concedes in detail. Epistemic logic [Hin62, FHMV95] indexes knowledge by agents, but it is modal and propositional: it models attitudes toward propositions in possible-worlds structures, not an extensional classification of the artifacts themselves. Awareness and unawareness logics [FH87, HMS06] formalize what an agent does not even conceive of, yet again without a functional axis that could distinguish, within the unaware domain, capacities that objectively operate from mere absence. Classifying *artifacts* rather than propositions is not itself a departure: object-based unawareness is an established line (Section 9.1). The ability axis likewise has ancestors: that what an agent can do is a matter distinct from what truths it knows is Ryle's distinction [Ryl49, Ryl46], it has a formal semantics of its own in the logics of knowing how [Wan15], and it appears as a primitive modality in the logics of strategic ability [AHK02]. What is specific here is the *form* the axis takes: functionality is predicated of the artifacts rather than of the agents, extensionally rather than modally, and jointly with two semantic predicates on the same domain, so that it cuts a classification into cells instead of qualifying a formula. We are not aware of a framework that jointly (i) carries a functionality predicate of that artifact-level, extensional kind alongside a semantic pair, (ii) obtains agent relativization parametrically rather than modally, and (iii) recovers, explicitly and per agent, the cell it excludes. This is

a statement about our coverage of the literature, not a claim that no such framework exists; the full comparison, with the same boundary, is developed in Section 9.1.

1.3. **Approach.** The question this paper asks is what such a classification determines. Three predicates over a domain induce, for each agent, a finite datum: which of the resulting cells are inhabited for it. That datum is a severe compression of the agent's state – it records occupancy and discards which artifacts do the occupying, what the agent can compose, and what it refers to. The paper studies both directions of that compression: what the finite datum still fixes, and what passing to it irretrievably loses.

The MCK framework starts from three independent binary predicates on an artifact domain $\mathcal{A}$: $M(x)$ holds when x is present as structured semantic content, integrated into the system's semantic organization; $C(x)$ holds when x is apprehended but not semantically integrated – recognized, yet opaque; $K(x)$ holds when x has operator-like, transformative character. M and C form a semantic axis on which they may overlap (partial integration); K is a functionality axis independent of both. The $2^3 - 1 = 7$ non-empty Boolean combinations of the predicates generate seven regions $R_1, \ldots, R_7$ that provably partition the classified domain (Theorem 4.8): pure understanding, pure opacity, partial integration, transparent operators, opaque operators, and the full triple each receive a definite cell, and the pure-K cell $(0, 0, 1)$ – functional yet semantically absent – is operationally suspended by an explicit convention rather than silently dropped (Axiom 4.11).

A further layer completes the static classification: it is *agent-relative*. Instantiating the predicate triple per agent, with no new primitives and no new axioms, yields region non-invariance (Lemma 5.20) – the same artifact occupies different regions for different agents – and a distinction between subjective and objective readings of K (Definition 5.14), under which the suspended pure-K cell is explicitly recovered per agent (Remark 5.19) as the region of *functional potential* (Definition 5.12): operational capacity that genuinely exists in the shared domain but lies outside a particular agent's semantic horizon, and whose certification that definition places with an external witness.

Both directions of the compression admit exact answers, and together they are the paper's principal finding. On the positive side the finite type fixes more than an occupancy list: the six structural features it determines are not freely combinable, and which combinations occur is characterized exactly (Theorem 7.12). It also fixes one behavioural fact outright. Because only artifacts in Θ denote, and because every question an agent poses is itself one of its symbols, a query requires a non-empty carrier (Theorem 8.22) – so the types whose carrier is empty, of which there are exactly seventeen, can pose no question whatever architecture they are given (Corollary 8.23). The implication runs in that direction only: a non-empty carrier does not by itself supply a query. On the negative side the compression is genuinely lossy, and provably so. Two agents may carry the same extensions, hence the same type, and still differ in what they can compose and in what they have recorded (Theorem 8.30). A finite occupancy type is therefore an exact instrument for some questions and no instrument at all for others, and the paper's task is to say which are which.

The paper is definitional and foundational. Its epistemic discipline is carried by the theorem environments and by the statements themselves: an *Axiom* is a framework stipulation, a *Theorem*, *Proposition*, *Lemma* or *Corollary* is proved from the definitions and axioms unless explicitly marked as a convention or as witnessed, and a *Remark* takes its status from its own text rather than from its environment: remarks below carry proven, witnessed, definitional, conjectural and expository content, the conjectural ones being stated as

conjectures and the proved ones carrying their argument. The proven core stays close to the definitions: the partition theorem (Theorem 4.8) with its relativized and derived forms, the predicate-toggle symmetry that partition carries (Proposition 4.20), non-invariance across agents (Lemma 5.20 and Remark 6.2), and the combinatorial and compositional facts these determine. The three independence stipulations are axioms, the R_3 suspension is a convention stated as an axiom (Axiom 4.11), one further axiom – output apprehension (Axiom 8.14) – is stipulated by the execution layer, and the satisfiability of functional potential is witnessed by construction rather than proven from the axioms (Lemma 5.22). The theory is static and extensional throughout: it classifies artifacts at a fixed semantic state, with no possible worlds, no accessibility relations, and no temporal structure. The two further layers it does develop are static as well. The compositional layer on the functionality axis is *static-generative*: it stratifies the functional set by derivation order under a fixed composition operation (Definitions 8.1 to 8.3), introducing no execution and no temporal structure. The execution layer (Section 8) is a relation on a finite set of region assignments together with its reflexive-transitive closure (Definition 8.12), carrying no time index and no cost model; the finite sequences of generation steps quantified over in Theorem 8.32 are derivation orders in the sense of Definition 8.3, not temporal runs. Region dynamics and the taxonomy of symbol kinds are explicitly downstream, developed in companion work (Section 10).

1.4. **Contributions.** One sentence states the paper's spine. Three independent predicates over an artifact domain, read per observer, carve a cube of eight cells – seven of them the regions of the classified domain, the eighth the absolute exterior, which the framework keeps in view rather than drops – and everything that follows studies what the finite pattern of occupied cells does and does not fix about the agent that occupies them.

Two *modelling choices* shape that study, and they are stipulations rather than results. The first is that the two semantic predicates are not required to exclude one another, so that $M \cap C$ may be inhabited (Axiom 3.11). The second is that denotation is restricted to that overlap: only artifacts in $\Theta = M \cap C$ denote (Definitions 4.16 and 4.17). Neither is proved here, and neither is offered as a discovery. What is proved is what follows from them.

- **C1 – Region structure and a finite type space.** The seven regions partition the classified domain (Theorem 4.8), the partition carries a predicate-toggle symmetry (Proposition 4.20), and relativizing the triple per agent makes region membership provably non-invariant across agents (Lemma 5.20). An agent's *type* is the set of cells it inhabits (Definition 7.1); exactly 109 of the 128 patterns are realized by an agent whose three extensions are all non-empty (Theorem 7.2). Those patterns admit a shorter description, and the paper gives it rather than presenting the count as new: reading each cell as the set of predicates that hold on it, the realized types are precisely the covers of $\{M, C, K\}$, so the two counts are the labelled and unlabelled cover numbers of a three-element set (Remark 7.3) [Com74, BLL98]. The contribution here is the identification – an agent's semantic–functional state is a cover of its own predicate set – and what the later sections extract from it.
- **C2 – Symbolic carrier, and a query obstruction that no architecture removes.** An agent is symbol-free – its denotation function has empty domain – exactly when cells 4 and 7 are absent from its type, which holds for exactly 17 of the 109 types (Corollary 7.5). Every query an agent poses is itself one of its symbols, so a query requires a non-empty carrier (Theorem 8.22); the implication runs in that direction only. Consequently those 17

types admit no query *under every compatible architecture* (Corollary 8.23): the obstruction is fixed by the type and cannot be lifted by architectural means. The individual steps here are small, and the paper says so where it makes them (Remark 8.24): that a query's token denotes, and that denotation is confined to the overlap, are stipulations, and the criterion for an empty carrier unfolds the definition of that overlap. What the group contributes is the composition – a datum recording only which cells an agent occupies settles an operational question about it under every architecture whatsoever – together with the finite count that makes the class of such agents explicit.

- **C3 – An exactly characterized space of structural profiles.** The coarser datum that the type alone fixes – whether an agent bears a symbol, an operable symbol, a transparent or an opaque operator, an inhabited generative cell, inert content – is a map Γ into $\{0,1\}^6$ (Definition 7.9). Its image is not merely enumerated but *characterized*: a profile is realized if and only if it satisfies four explicit Boolean constraints (Theorem 7.12), from which $|\operatorname{Im}\Gamma| = 39$ of 64 follows (Corollary 7.13) and which yield, by the same route, that the realized profiles are closed under coordinatewise join but not under meet, forming a finite join-semilattice with a greatest and no least element (Proposition 7.14). An exhaustive computation over the 109 types verifies the finite claims independently of the proofs.
- **C4 – What a summary of the outcome fails to fix.** The sharpest form of this is not about the type at all, and concerns a far richer summary. The generative layer converges: internal generation reaches the compositional closure of its basis and stops there (Theorem 8.32). That closure is the whole set of artifacts the architecture produces, and it does not determine how they were produced – there is no function recovering, from the closure alone, either the generation depth of its elements or the pairs on which the composition is defined (Corollary 8.8), and the hypothesis under which this holds is exactly sharp (Remark 8.9). A coarser consequence of the same construction concerns the type: two agents may carry the same complete triple, hence the same type, while their architectures differ, and then their execution and linkage memories differ (Theorem 8.30); neither is a function of the type. The paper separates four kinds of dependency accordingly – what the type fixes, what only the extensions fix (storage, and Θ_a as a set), what the architecture is sensitive to (execution), and what needs more than either (linkage, and the availability of a query) – and does not collapse them. Against that negative result stands a positive one on the architecture side: with the architecture held fixed, internal generation reaches its fixed point at the compositional closure of the basis, after exactly the stabilization depth and not fewer iterations, and admits no further step (Theorem 8.32). The step relation carries no time index and no cost model, and contains no operation of supply, so it models no acquisition.

Scope. The predicates are supplied model data. Every result is a statement about an arbitrary assignment meeting its hypotheses; identifying the intended assignment from external evidence is outside the present formalism, and Section 10 records what that leaves open.

Non-claims. To fix the scope, the paper does *not* claim: a dynamics of region transitions – no cost, no trajectory, no adjacency topology on the regions (what is established is the static partition, its symmetry, and the coherence conditions on (source, transmutation) pairs); a metric or embedding geometry on the artifact domain; a proof, from the axioms, of the K-objective satisfiability of functional potential (it is witnessed); or novelty of any single ingredient (tripartite approximation, agent indexing, unawareness in both its state-space

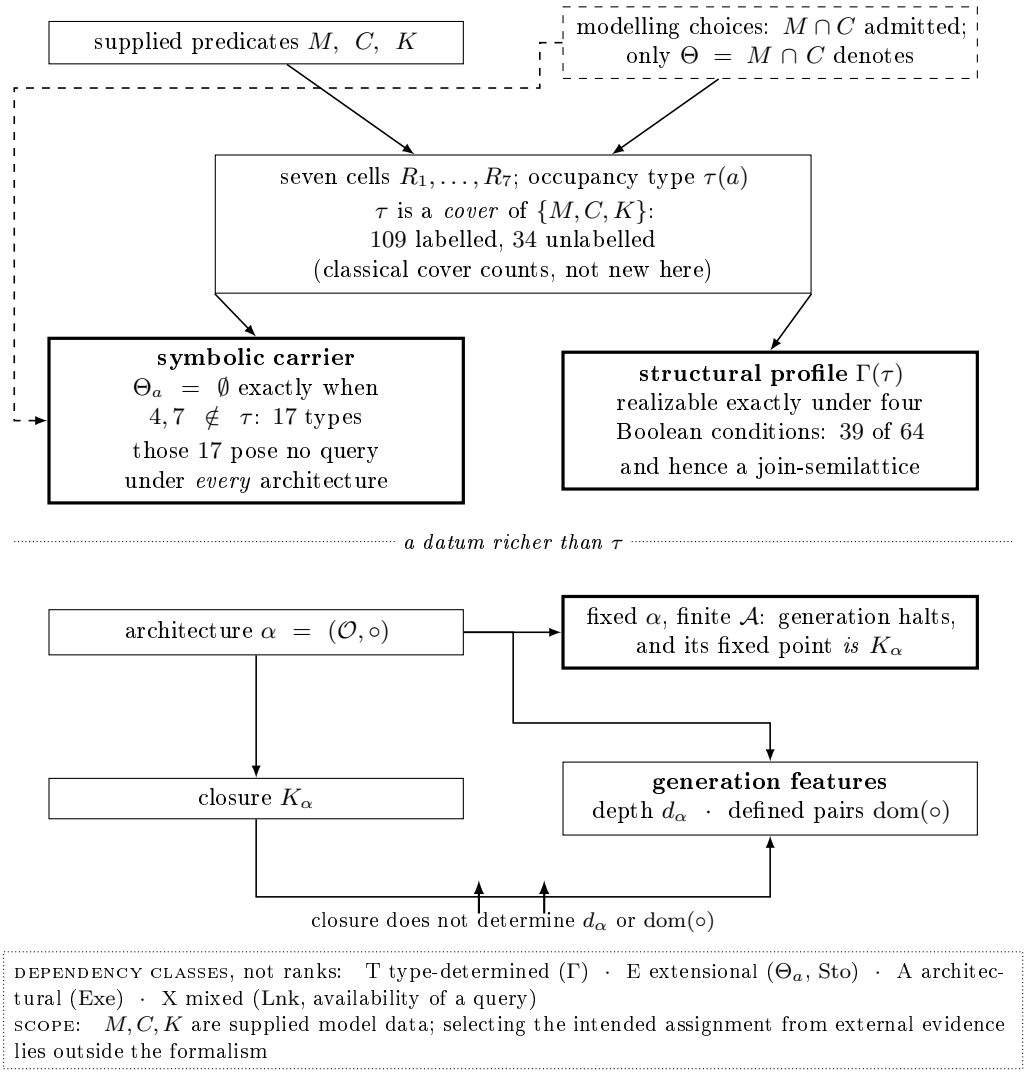


Figure 1: **Dependency map of the framework.** Solid arrows are formal dependency or construction. The dashed arrow is a *definitional* bridge: it holds by unfolding a definition rather than by proof, and the obstruction it feeds is proved. The connector labelled *does not determine* records a proved *non-determination*. Boxes may group several separately defined objects for presentation only, adding no relation: that single connector abbreviates the two conclusions of one corollary – that no function of the closure recovers the generation depth, and none recovers the domain of the composition – a phenomenon first possible once the functional set has two elements. The counts 109 and 34 are classical cover numbers and are not claimed here. T, E, A and X are dependency classes, not ranks.

and its object-based form, Boolean attribute lattices, generated subalgebras, and the ability axis are all established – the contribution is their combination, Section 9.1).

Organization. The paper follows the dependency order of its objects. The observer motivation (Section 2) is informal and no result rests on it; the predicate layer (Section 3) fixes M, C and K and the sense in which they are supplied; the region theory (Section 4) proves the partition and introduces Θ and denotation; the multi-agent layer (Section 5) relativizes the triple and proves non-invariance; Section 6 names the semantic horizon. Section 7 classifies the types, characterizes the symbol-free ones, and isolates the structural profile Γ. Section 8 supplies the architecture and closure machinery, defines the step relation and the query, proves the query horizon and the separation of execution and linkage from type, and locates the fixed point of internal generation. Sections 9 and 10 position the framework and record what it leaves open; Section 11 concludes.

2. The Agentic Observer

The three primitive predicates of the MCK framework are not arbitrary coordinates imposed on the artifact domain. Before they are posited, in Section 3, as independent binary predicates, they can be motivated from a single, more elementary starting point: the situation of an *agentic observer* that confronts artifacts and must come to terms with them. This section carries out that motivation informally. It proves nothing; its purpose is to make the axioms of Section 3 read as the formalization of a motivating account rather than as three stipulations introduced cold.

One capacity, two outcomes. An agentic observer that apprehends an artifact has one basic capacity with respect to it: to *understand* it, or to *fail to understand* it. Everything semantic in the framework differentiates from this single capacity. What the observer apprehends *and* understands is present to it as structured meaning: its content is organized, its relational position among other artifacts is stable, its behavior under known operations is predictable. This is the condition fixed in Section 3 as the predicate M. What the observer apprehends but does *not* understand is present without being integrated: recognized, named, perhaps even invoked, yet opaque to it. This is the condition fixed as the predicate C. The two semantic predicates are thus the two outcomes of one capacity applied to what is apprehended – the understood and the merely registered – not two independent primitives that happen to coexist. They need not be exclusive: an artifact can be partly understood and partly opaque, and this overlap is exactly the structured ambiguity domain $\Theta = M \cap C$ developed in Section 4.5.

A third, independent axis. Understanding does not exhaust what an observer can register about an artifact. Independently of whether the artifact is understood or opaque, it may *do* something: it may act on other artifacts and produce outputs, or it may be inert. This operative potential is a third axis, orthogonal to the semantic one – present, in principle, even where meaning and opacity are both absent. It is fixed as the functionality predicate K. That this axis is genuinely independent of the semantic pair, rather than reducible to it, is the content of the framework's independence stipulations (Axioms 3.10 to 3.12), which the following section states as axioms.

Functionality as originating potential. The cell of pure functionality – operative potential with neither meaning nor opacity, $K = 1$ and $M = C = 0$ – occupies a distinguished place in this differentiation. It is the region R_3 (Definition 4.3), the *generative unknown*: functional force that has not yet entered the observer's semantic horizon and so has not yet differentiated into anything understood or even recognized. In the static, single-observer setting of this paper R_3 is operationally suspended by convention (Axiom 4.11); read positively, however, it is the originating potential from which classification has not yet separated determinate structure. It is the trace, within the fully formed region cube, of the point at which the observer's apprehension has an operative charge but no semantic differentiation of it has yet occurred.

Treating a reservoir of undifferentiated generative potential as the origin from which determinate structure separates out has a long intellectual precedent: the classical notion of an originating principle, an *archē* – in the doxographic tradition that reaches us through Aristotle and Theophrastus, Anaximander's boundless (*apeiron*), the source from which determinate things separate out [KRS83]. Both attributions carry caveats: whether *archē* was Anaximander's own term in this technical sense is disputed, the label being supplied by the later tradition, and the sense of *apeiron* – spatially unlimited, internally indeterminate, or qualitatively unbounded – is likewise contested. We invoke the analogy only to motivate reading functionality as a generative origin rather than as a derived attribute; nothing in the formal development depends on it, and nothing here is offered as a reading of the sources.

Handoff to the formal development. This informal derivation fixes the intended reading of the three capacities but establishes no result, and no formal treatment of the derivation itself is attempted here. We now fix these three capacities as the primitive predicates of the framework – M and C as the two outcomes of the observer's capacity to understand what it apprehends, and K as the independent axis of operative potential – and their mutual independence as axioms, from which the seven-region structure follows as a theorem. The remainder of the paper is the formal development of what this section has only motivated.

3. Foundational Predicates

The three predicates defined below formalize the three capacities of the agentic observer motivated in Section 2: the two outcomes M and C of its capacity to understand what it apprehends, and the independent functionality axis K. They are posited here as primitives; the informal genesis of Section 2 is not part of the formal development.

Scope and conventions. This section and Section 4 develop the static core of the MCK framework. All results are *static classification* results: they classify artifacts at a fixed semantic state and assert no temporal properties. No claim in this paper quantifies over execution traces, so no temporal (LTL) schema is attached to any claim. The epistemic status of every non-trivial claim – definitional, proven, witnessed, or conjectured – is stated in the surrounding prose.

The predicates are *supplied* rather than computed, and this fixes the reach of everything below. Each result is a statement about an arbitrary assignment satisfying the conditions its own hypotheses name: the framework takes M, C and K as model data and derives from them the region structure, the classification of agent types, and their consequences.

Selecting an intended assignment, or validating one against external evidence, is outside the present formalism; Section 10 records what that leaves open.

3.1. **Primitive Definitions.**

Definition 3.1 (Artifact domain $\mathcal{A}$)**.** Let $\mathcal{A}$ denote the domain of discourse: the totality of artifacts under consideration. An artifact is any discrete entity – a concept, a datum, an operator, a symbol, a system – that may be classified within the semantic-functional space of the framework.

Definition 3.2 (Predicate M – structured meaning)**.** For every artifact $x \in \mathcal{A}$ (Definition 3.1), the predicate $M(x) \in \{0, 1\}$ indicates whether x is present as *structured semantic content*. When $M(x) = 1$, the system possesses a constructive witness – a certificate of understanding – demonstrating that x has been integrated into the semantic manifold: over the aspects of x that the certificate covers, its relational position among other artifacts is stable, its properties are enumerable, and its behavior under known operations is predictable. The certificate is not required to be exhaustive: $M(x) = 1$ asserts that such a witness *exists*, not that every aspect of x has been resolved by it. Whether an unresolved remainder is left over is the separate question recorded by C (Definition 3.3), and the two predicates are correspondingly not complementary (Axiom 3.11).

Definition 3.3 (Predicate C – opacity)**.** For every artifact $x \in \mathcal{A}$ (Definition 3.1), the predicate $C(x) \in \{0, 1\}$ indicates whether x is *apprehended but not fully semantically integrated*. When $C(x) = 1$, the system recognizes x as an artifact – it is registered in the ontology – but its resolution of x is *incomplete*: at least one aspect of x's internal structure remains unresolved, so the system's certificate for x, if it has one at all, does not cover x in full. $C(x) = 1$ therefore records a *residual* opacity, not the total absence of understanding: the limiting case in which no aspect at all is resolved is the conjunction $C(x) = 1 \wedge M(x) = 0$, and the case in which some aspects are certified while a remainder is not is $C(x) = 1 \wedge M(x) = 1$ (Axiom 3.11). In either case the unresolved remainder may carry latent information that, under the right operations, can be extracted.

Definition 3.4 (Predicate K – functionality)**.** For every artifact $x \in \mathcal{A}$ (Definition 3.1), the predicate $K(x) \in \{0, 1\}$ indicates whether x possesses *operator-like or transformative character*. When $K(x) = 1$, the artifact participates in structured mappings: it can act on other artifacts to produce outputs. K captures the dimension of generativity – the capacity of x to participate in or define transformations.

The three predicates are primitive and jointly characterize the semantic-functional status of every artifact:

$$M(x),\ C(x),\ K(x) \in \{0, 1\} \quad \text{for all } x \in \mathcal{A}.$$

Notational convention. We write M, C, K both for the predicates and for their extensions $\{x \in \mathcal{A} \mid M(x) = 1\}$, etc.; set-theoretic expressions such as $M \cup C$ and $M \cap C$ are always read over extensions.

3.2. **Interpretation.**

$M(x)$ – **structured semantic content.** $M(x) = 1$ is not generic "interpretability": it marks the presence of a data-structure-like semantic object. The *certificate of understanding* witnessing it demonstrates three things – a stable relational position among other artifacts, enumerable and verifiable properties, and predictable behaviour under known operations – over the aspects of x it covers. Artifacts in M are the ordered, low-entropy portion of the semantic domain.

The certificate is not free. Three scope statements about Definition 3.2 are owed here, and each narrows it. First, the word *certificate* names the three conditions the definition lists and nothing further. It is not an algorithm, not a proof object in a deductive system, and not a procedure by which the framework decides membership in M: the predicates are supplied, and the word marks what a supplied assignment is intended to record. Second, the word must be read as constrained, and specifically as carrying the three conditions just named. It must not be read as satisfied by any exhibition of x as an output whatever: under that unrestricted reading the predicate is vacuous, since for a suitable choice of operator and operands any artifact at all can be displayed as the value of a known function on known inputs, so M would hold everywhere and separate nothing. This paper does not supply the constraint that makes the intended reading precise; it states only that the intended reading is the constrained one, and records the gap as a limitation (Section 10). Third, the evidence that bears on the predicate is asymmetric. A certificate exhibited establishes $M(x) = 1$. A search that fails to produce one does not establish $M(x) = 0$; it establishes that none was produced by that search. Assignments of $M(x) = 0$ are therefore relative to the search behind them, and every region membership whose defining conjunction contains $M(x) = 0$ – R_2, R_3 and R_6 (Definitions 4.1 to 4.7) – inherits that relativity.

$C(x)$ – **opacity.** $C(x) = 1$ asserts that x is apprehended – recognized as existing – but not *fully* semantically integrated: some aspect of its internal structure is left opaque, so no certificate of understanding covers x in full. C covers unresolved symbolic reference and representation-as-symbol. Artifacts in C are *mineable*: under appropriate operations their residual latent structure may be extracted, resolving the remainder and, in the limit, removing them from C.

$K(x)$ – **functionality.** $K(x) = 1$ asserts operator-like character: x participates in structured mappings and can act on other artifacts to produce outputs. K is orthogonal to the semantic axis – an artifact may be functional whether understood or opaque – and encodes the potential for structured action, not semantic content: $K(x) = 1$ does not mean x is "known".

3.3. **Derived Sets.**

Definition 3.5 (Known and unknown domains)**.** The *known domain* and the *unknown domain* are

$$\text{KNOWN} = M \cup C, \qquad \text{UNKNOWN} = \{x \in \mathcal{A} \mid M(x) = 0 \wedge C(x) = 0\}.$$

All semantically present artifacts – whether understood or opaque – belong to the known domain (Definitions 3.2 and 3.3); KNOWN is a *derived* set, not a primitive, defined from M and C. Artifacts in the unknown domain lie outside semantic integration entirely: the system holds no certificate of understanding for them (Definition 3.2) and registers no unresolved

remainder in them (Definition 3.3), so it records nothing semantic about them at all and they are beyond the current semantic horizon.

Proposition 3.6 (Known/unknown partition)**.** KNOWN *and* UNKNOWN *partition* $\mathcal{A}$*:*

$$\text{KNOWN} \cup \text{UNKNOWN} = \mathcal{A}, \qquad \text{KNOWN} \cap \text{UNKNOWN} = \emptyset.$$

Proof. For any $x \in \mathcal{A}$, either $M(x) = 1$ or $C(x) = 1$ (placing x in KNOWN), or both $M(x) = 0$ and $C(x) = 0$ (placing x in UNKNOWN). These conditions are exhaustive and mutually exclusive. □

Definition 3.7 (Classified domain)**.** The *classified domain* is

$$\mathcal{A}^+ = \{x \in \mathcal{A} \mid M(x) \vee C(x) \vee K(x)\}.$$

An artifact belongs to the classified domain if and only if at least one of the three predicates (Definitions 3.2 to 3.4) holds. Every artifact in $\mathcal{A}^+$ is assigned to exactly one of the seven MCK regions (proven in Section 4.2). The complement $\mathcal{A} \setminus \mathcal{A}^+$ consists of artifacts for which no predicate holds; these lie in the *absolute exterior* of the framework.

Proposition 3.8 (Stratification of the unknown domain)**.** *The unknown domain decomposes as*

$$\text{UNKNOWN} = \{x \mid M(x) = 0,\ C(x) = 0,\ K(x) = 1\} \ \cup\ (\mathcal{A} \setminus \mathcal{A}^+),$$

a disjoint union of the pure-functionality cell *– functionally active but semantically absent artifacts, named R_3 in Definition 4.3 – and the artifacts for which no predicate holds.*

Proof. Let $x \in$ UNKNOWN. Then $M(x) = 0$ and $C(x) = 0$. Either $K(x) = 1$, placing x in the pure-functionality cell, or $K(x) = 0$, placing x in $\mathcal{A} \setminus \mathcal{A}^+$. The two cases are exhaustive and mutually exclusive. □

Remark 3.9 (The absolute exterior). The passage from $\mathcal{A}$ to $\mathcal{A}^+$ excludes exactly the $(0, 0, 0)$ predicate cell – the absolute exterior. This excluded configuration is acknowledged explicitly rather than silently dropped: Definition 3.7 names it, and Proposition 3.8 locates it inside UNKNOWN. The distinction between the absolute exterior – artifacts absent from the ontology – and the registered-but-opaque artifacts of C mirrors the open- versus closed-world contrast: the framework adopts an open-world reading, in which an unregistered artifact is genuinely absent rather than assumed false as under the closed-world assumption [Rei78]. No result in this paper quantifies over the absolute exterior beyond Proposition 3.8.

3.4. **Axis Structure.** The three predicates define two orthogonal axes of classification.

Axis 1: semantic / entropy axis. Defined by M and C. This axis measures the degree of semantic integration of an artifact: M corresponds to low entropy – structured, integrated, stable semantic content; C corresponds to high entropy – opaque, unstructured, potentially mineable presence. The predicates M and C are not mutually exclusive: an artifact may satisfy both $M(x) = 1$ and $C(x) = 1$, indicating partial integration with residual opacity.

Axis 2: functionality axis. Defined by K. This axis measures the generative capacity of an artifact: $K = 1$ means the artifact has transformative, operator-like behavior; $K = 0$ means the artifact is inert with respect to transformations.

The three predicates are *stipulated* to be mutually unconstrained. This is the framework's founding commitment, and it is posited rather than derived; we state it as three axioms. The Axiom/Theorem split is the epistemic signal of this paper: an Axiom is a framework stipulation, a Theorem or Proposition is proved from the definitions and axioms.

Axiom 3.10 (Axis independence)**.** The functionality axis is independent of the semantic axis: an artifact can be understood ($M = 1$) but non-functional ($K = 0$), or functional ($K = 1$) but opaque ($C = 1$, $M = 0$).

Axiom 3.11 (Non-exclusivity of M and C)**.** The framework imposes no exclusion between M and C; their extensions may overlap, $M \cap C \neq \emptyset$ (admissible, not asserted). Artifacts in $M \cap C$ are partially integrated: some aspects are understood while residual opacity remains – the structured ambiguity domain developed in Section 4.5.

Axiom 3.12 (Independence of K)**.** K is logically independent of both M and C; no implication holds between K and either of them:

$$M(x) \not\Rightarrow K(x), \quad K(x) \not\Rightarrow M(x), \quad C(x) \not\Rightarrow K(x), \quad K(x) \not\Rightarrow C(x).$$

The content of the axiom is exactly these four displayed non-implications. It denies no other sentence in the predicate language; in particular it does not deny the disjunctive $K(x) \Rightarrow M(x) \vee C(x)$, which is strictly weaker than $K \Rightarrow M$ and $K \Rightarrow C$ and which the R_3 suspension (Axiom 4.11) and the subjective reading of K (Definition 5.14) each impose over their own scope.

Remark 3.13 (Intuition). The three axioms say the same thing from three angles: Definitions 3.2 to 3.4 impose no implication among the predicates, so all eight predicate triples $(M(x), C(x), K(x))$ are admissible a priori. It is precisely the independence of K from M and C that generates the seven-region decomposition of Section 4, in which each KNOWN artifact is classified by its unique triple (Theorem 4.8, Corollary 4.10).

Remark 3.14 (Graded and information-theoretic refinements: deferred). The predicates of Definitions 3.2 to 3.4 are binary and stipulated. A graded refinement – one grounding membership in M, C, or K in a quantity admitting degrees, with a notion of distance to a region and whatever structure that would induce – is not developed here, and we do not develop it elsewhere either: the present paper introduces no metric primitive, and every result below uses only the binary predicates. The graded reading is named at this point so that a reader does not mistake the binary predicates for a claim that the underlying distinctions admit no degrees; it is an open direction, not a deferred result.

4. Formal MCK Regions

The three binary predicates M, C, K (Definitions 3.2 to 3.4) assign to each artifact in $\text{KNOWN} \cup \{x \mid K(x) = 1\}$ – that is, to each artifact in the classified domain $\mathcal{A}^+$ (Definition 3.7) – a unique triple $(M(x), C(x), K(x))$. Excluding the empty triple $(0, 0, 0)$, which corresponds to the absolute exterior, there are $2^3 - 1 = 7$ possible non-trivial combinations. Each combination defines a region.

4.1. **Region Definitions.**

Definition 4.1 (Region R_1 – pure understanding)**.**

$$R_1 = M \setminus (C \cup K) = \{x \mid M(x) = 1,\ C(x) = 0,\ K(x) = 0\}.$$

Definition 4.2 (Region R_2 – pure opacity)**.**

$$R_2 = C \setminus (M \cup K) = \{x \mid M(x) = 0,\ C(x) = 1,\ K(x) = 0\}.$$

Definition 4.3 (Region R_3 – generative unknown)**.**

$$R_3 = K \setminus (M \cup C) = \{x \mid M(x) = 0,\ C(x) = 0,\ K(x) = 1\}.$$

Instantiation in this cell is operationally suspended by an explicit convention (Axiom 4.11).

Definition 4.4 (Region R_4 – partial integration)**.**

$$R_4 = (M \cap C) \setminus K = \{x \mid M(x) = 1,\ C(x) = 1,\ K(x) = 0\}.$$

Definition 4.5 (Region R_5 – transparent operator)**.**

$$R_5 = (M \cap K) \setminus C = \{x \mid M(x) = 1,\ C(x) = 0,\ K(x) = 1\}.$$

Definition 4.6 (Region R_6 – opaque operator)**.**

$$R_6 = (C \cap K) \setminus M = \{x \mid M(x) = 0,\ C(x) = 1,\ K(x) = 1\}.$$

Definition 4.7 (Region R_7 – full triple)**.**

$$R_7 = M \cap C \cap K = \{x \mid M(x) = 1,\ C(x) = 1,\ K(x) = 1\}.$$

4.2. **Mutual Exclusion and Exhaustiveness.**

Theorem 4.8 (Seven-region partition)**.** *The seven regions $R_1, \ldots, R_7$ partition the classified domain $\mathcal{A}^+ = \{x \in \mathcal{A} \mid M(x) \vee C(x) \vee K(x)\}$.*

Proof. Pairwise disjointness. Each region corresponds to a distinct predicate triple, drawn from the seven elements of $\{0,1\}^3$ other than the empty one. Since the triple is unique for each artifact, the regions are pairwise disjoint: $R_i \cap R_j = \emptyset$ for $i \neq j$.

Exhaustiveness. For any artifact x with at least one predicate equal to 1, the triple $(M(x), C(x), K(x))$ matches exactly one of the seven non-zero elements of $\{0,1\}^3$. Therefore

$$R_1 \cup R_2 \cup R_3 \cup R_4 \cup R_5 \cup R_6 \cup R_7 = \mathcal{A}^+.$$

The complement $\mathcal{A} \setminus \mathcal{A}^+ = \{x \mid M(x) = 0,\ C(x) = 0,\ K(x) = 0\}$ is the absolute exterior. It is a proper subset of UNKNOWN, which additionally contains R_3 (Proposition 3.8). □

Figure 2 visualizes the partition as the seven non-empty zones of the $M/C/K$ Venn diagram, with the two excluded predicate cells marked explicitly.

Remark 4.9 (What the picture is faithful to). The framework defines no map from artifacts to the plane. Definition 5.1 supplies three subsets of $\mathcal{A}$ and nothing else: no artifact is located at a point, and M, C, K carry no proximity, adjacency, order or path. A drawing is therefore faithful *exactly* at the level of which cells are non-empty, and at no finer level. No point of Fig. 2 has an MCK property – not because the property has not been worked out, but because there is no artifact for the point to be about. Connectedness, tangency, incidence, area, adjacency and betweenness are structure the rendering possesses and the classification does not. This specifies the illustration rather than faulting it: what a picture

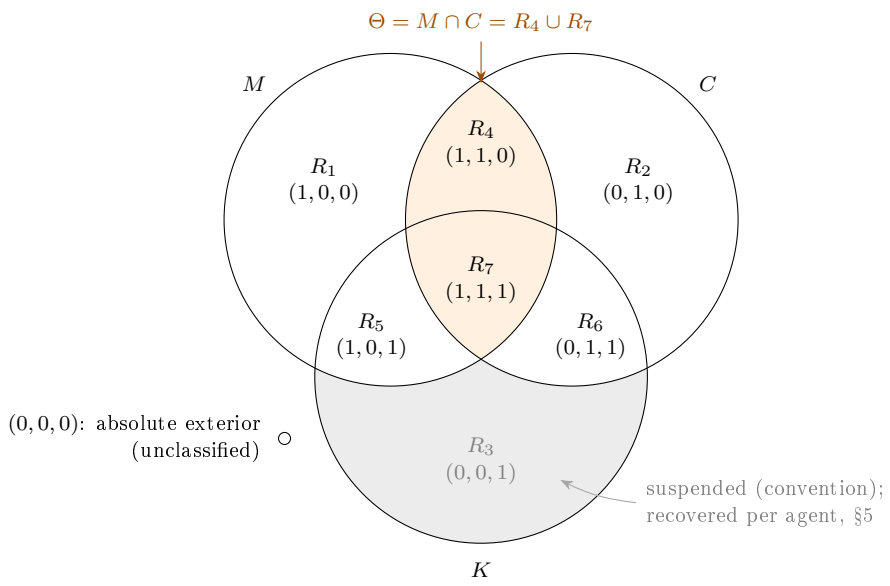


Figure 2: The $M/C/K$ Venn diagram. Its seven non-empty zones are the regions $R_1, \ldots, R_7$ (Definitions 4.1 to 4.7), which partition the classified domain $\mathcal{A}^+$ (Theorem 4.8); each zone is annotated with its predicate triple $(M(x), C(x), K(x))$. $R_7 = M \cap C \cap K$ is the triple intersection at the center. The greyed pure-K crescent $R_3 = (0, 0, 1)$ is operationally suspended by convention (Axiom 4.11) and recovered per agent in Section 5 (Remark 5.19). The region outside all three circles is the absolute exterior $(0, 0, 0)$, outside $\mathcal{A}^+$ (Remark 3.9). The tinted lens is the structured ambiguity domain $\Theta = M \cap C = R_4 \cup R_7$ (Definition 4.16). Which cells are adjacent to which – the combinatorial structure this partition carries – is drawn separately in Fig. 3.

licenses one to say is which cells are inhabited, and nothing else. Every result of this paper is a statement about that datum.

Corollary 4.10 (Region uniqueness)**.** *Every semantically present artifact ($x \in$ KNOWN) belongs to exactly one of the seven regions.*

Proof. KNOWN $\subseteq \mathcal{A}^+$, since $M(x) \vee C(x)$ implies $M(x) \vee C(x) \vee K(x)$. Apply Theorem 4.8. □

4.3. R_3 Suspension.

Axiom 4.11 (R_3 suspension; scoping convention)**.** Pure $R_3 = K \setminus (M \cup C)$ (Definition 4.3) is formally defined but, *within the static, agent-independent classification developed in this paper*, operationally suspended from node instantiation: we stipulate, as a scoping convention, that no artifact is instantiated in R_3. Like the independence stipulations of Section 3.4, this is a posited framework commitment and not a derived statement, which is why it is stated as an axiom; unlike them, what it constrains is which artifacts are instantiated rather than what the primitive predicates imply of one another, so it is an axiom of the theory rather than a condition on its models. It is a scoping choice of the present exploration, not a

claim that R_3 is empty in general – Section 5 recovers it per agent, and its positive content is discussed in Remark 4.12.

Remark 4.12 (Justification of the suspension). The suspension is a justified framework *convention*, not a theorem. An artifact with $K = 1$, $M = 0$, $C = 0$ would be functional yet neither understood nor recognized; in the single classifier's static setting K presupposes at least semantic presence ($M = 1 \vee C = 1$), since a transformation that is not semantically present at all cannot be observed, verified, or classified as functional. R_3 has a *positive*, dual reading as the region of *generative unknown*: generative (it carries the functional force $K = 1$) yet unknown (outside the agent's semantic body, $\neg M \wedge \neg C$). It names latent functional capacity that may crystallize when the artifact enters the semantic horizon; suspension removes it from instantiation in this paper's static classification without denying that potential, whose quantitative characterization is downstream work.

Remark 4.13 (Forward pointer: agent-relative reading). Under the agent-relativized, K-subjective reading of the predicates (developed in Section 5 as Definition 5.14), the emptiness of the agent-relative region R_3^a becomes a provable statement rather than a convention. No such proof is claimed in this section; in the agent-independent setting of the present section the suspension remains a convention, and Axiom 4.11 is stated accordingly. The recovery of this excluded cell is carried out in Remark 5.19.

Consequence. The active partition of KNOWN uses six regions: R_1, R_2, R_4, R_5, R_6, R_7. All references to "the MCK regions" in subsequent sections denote these six active regions unless stated otherwise.

Remark 4.14 (The suspended cell). The suspension excludes the $(0, 0, 1)$ predicate cell from operational instantiation while retaining it in the formal partition. This is an acknowledged excluded configuration: Theorem 4.8 remains a statement over all seven cells of $\mathcal{A}^+$, whereas operational claims about instantiated nodes range over the six active regions only. Together with the absolute exterior $(0, 0, 0)$ (Remark 3.9), these are the only two cells of $\{0, 1\}^3$ excluded from operational classification, and both exclusions are explicit.

4.4. **Semantic and Operational Meaning.** Each active region has a definite operational reading:

- R_1 – **Pure Understanding** $(1, 0, 0)$: a *stable referent*, fully integrated, no residual opacity, not functional – structured meaning in its purest form.
- R_2 – **Pure Opacity** $(0, 1, 0)$: a *mineable unknown*, recognized but neither understood nor functional; a candidate for promotion to M-status.
- R_4 – **Partial Integration** $(1, 1, 0)$: a partial certificate of understanding with residual opacity, not functional; transitional between R_1 (opacity resolves) and R_2 (understanding collapses).
- R_5 – **Transparent Operator** $(1, 0, 1)$: understood *and* functional – the framework's structured, executable tools.
- R_6 – **Opaque Operator** $(0, 1, 1)$: functional but not understood – operationally effective yet semantically opaque, the important black-box class.
- R_7 – **Full Triple** $(1, 1, 1)$: partially understood, partially opaque, and functional – a complex artifact the system can still operate and compose.

Remark 4.15 (K as a potential field: an interpretive reading, deferred). K is not attributed the way M and C are. One does not *mine* functionality as one mines opacity: it is a structural property of how an artifact relates to transformations, discovered by observing that x participates in structured mappings (Definition 3.4), not extracted from its content. The canonical framework reads the extension of K as a *potential field*: a reservoir of transformation capacity, dense where functional structure concentrates and predictive of where compositional emergence is productive. This is an interpretive reading, tagged as a conjecture: it adds no formal content to Definition 3.4, no result of this paper depends on it, and its quantitative development – density, gradient, and the geometry either would presuppose – is not carried out here and is not carried out elsewhere: no metric primitive is introduced in this paper, and the field reading remains a metaphor awaiting a formalism rather than a result held in reserve. The one exact precisification of "functional potential" available within this paper is the multi-agent construction of Section 5 (Definition 5.12), not the field metaphor.

4.5. **Structured Ambiguity Domain.**

Definition 4.16 (Structured ambiguity domain Θ)**.**

$$\Theta = M \cap C = R_4 \cup R_7.$$

Θ is the domain of *structured ambiguous knowledge*. It contains all artifacts for which both $M(x) = 1$ and $C(x) = 1$: meaning is present, but so is unresolved symbolic or opaque structure. The second equality follows from Definitions 4.4 and 4.7, since $M \cap C = ((M \cap C) \setminus K) \cup (M \cap C \cap K)$.

Θ is *structured* because it intersects M: there is genuine meaning-bearing structure present. It is *recognized* because it lies inside the known domain $M \cup C$. It is *unresolved* because it intersects C: the content retains symbolic, layered, or partially opaque character. Θ is not: pure ignorance (it contains structured meaning); pure meaning (it retains opacity); pure notation (it has semantic organization). It is the proper domain of symbols that carry meaning while remaining incompletely resolved: symbols live in Θ, not in pure M – pure R_1 artifacts are data structures, fully resolved and without symbolic character. The non-functional part of Θ is R_4; its functional part is R_7.

This placement rests on a dual-aspect reading that every apprehended artifact admits. Considered as a string, an artifact s carries a *meaning* aspect – it can be taken as a data structure, content to be read off directly – and a (possible) *symbol* aspect – it can be taken as a reference that points beyond itself to what it denotes. The two aspects are generically both available for a given s. Pure meaning is exhausted by the first aspect: an artifact in R_1 (Definition 4.1) is a fully resolved data structure, with no residual pointing. Symbolic character is precisely the second aspect, and it requires residual opacity: for s to refer beyond itself, part of it must remain unresolved, so that what it denotes is not identical to what it directly is. We adopt this requirement as a *stipulation* of the framework rather than derive it, and record it as such: it is the second of the two modeling choices on which the results below rest, the first being the non-exclusivity of M and C (Axiom 3.11). The long-standing practice of distinguishing a sign from its sense and from what it denotes, and of holding that a sign presents its object only in some respect, is the background against which the stipulation is a natural one; what this paper adds is its extensional form: the remainder that makes the pointing possible is the predicate C, and the cell in which the two aspects

coexist is Θ. A symbol therefore cannot inhabit pure M; it lives in $\Theta = M \cap C$, where structured meaning – the name it presents – coexists with unresolved reference – what the name points to. This is exactly why Definition 4.17 restricts denotes to Θ. The classical single-referent case is then a symbol whose target set (Remark 4.18) is the singleton of a fully resolved referent – an artifact with $C = 0$, for instance an R_1 artifact (Definition 4.1) – and the classification of symbols by the shape of $T(s)$ that this case opens is the taxonomy deferred there. The finer taxonomy of symbol kinds and the axioms of the denotation graph they induce are developed in the companion on embedding dynamics [Bra26].

The following definition records, in compact form, the static denotation apparatus attached to Θ.

Definition 4.17 (Denotation function)**.** Let Θ be as in Definition 4.16 and let Ref denote the codomain of *referents*. The denotation function

$$\text{denotes}\colon \Theta \to \text{Ref}$$

maps each artifact $s \in \Theta$ – a *symbol* – to a referent; every denotation target lies in the classified domain $\mathcal{A}^+$ (Definition 3.7), within the active regions (Section 4.3). The function is: (i) *static* – denotation does not change under the region dynamics treated in downstream work; (ii) *single-valued* – each application of denotes yields exactly one referent, the full set of targets of a symbol being collected by Remark 4.18; and (iii) *domain-restricted* – only Θ-artifacts denote: only symbols refer.

Remark 4.18 (Target set). The *target set* of a symbol $s \in \Theta$ (Definition 4.16) is $T(s) = T_{\text{direct}}(s) \cup T_{\text{indirect}}(s)$: the referents reachable from s by a single denotation edge, together with those reachable along directed denotation paths whose internal nodes are all symbolic (in Θ). The full denotation-graph axiom system and the taxonomy of canonical symbol classes built on $T(s)$ are outside this paper's scope; only denotes and $T(s)$ are needed here to fix the semantic role of Θ.

4.6. **Region Poset Structure.** The seven regions form a graded poset under predicate inclusion. The structure is that of the non-empty subsets of a three-element set:

- **Atomic regions** (single predicate true): R_1, R_2, R_3. Rank 1.
- **Binary regions** (exactly two predicates true): R_4, R_5, R_6. Rank 2.
- **Full triple** (all three predicates true): R_7. Rank 3.

Each transition from an atomic region to a binary region corresponds to the activation of one additional predicate; each transition from a binary region to R_7 corresponds to the activation of the third.

Remark 4.19 (Algebraic characterization). The region poset is order-isomorphic to $2^{\{M,C,K\}} \setminus \{\emptyset\}$, the poset of non-empty subsets of a three-element set ordered by inclusion; the isomorphism sends each region to the set of predicates active on it (Theorem 4.8). This is equivalently the Boolean lattice B_3 with its bottom element $\hat{0}$ removed (its set of non-minimum elements; note the top element R_7 is retained, so this is not the proper part $B_3 \setminus \{\hat{0}, \hat{1}\}$), and is isomorphic to the face poset of the 2-simplex (triangle) excluding the empty face. The structure is a finite bounded-above join-semilattice: every pair of regions has a join (predicate enrichment), the top element is R_7, but meets do not universally exist – the meet of two atomic regions with disjoint predicate support would be $\hat{0} = (0, 0, 0)$, which lies outside the classified domain $\mathcal{A}^+$. The rank function $\rho(R_i)$ equals the number of active predicates,

providing a natural complexity grading. This poset and its properties are instances of standard constructions in enumerative combinatorics; see Stanley [Sta12, Ch. 3] for the general theory of Boolean lattices and their rank-selected subposets.

Complementary to the order structure of Remark 4.19, the same cube carries a *symmetry* structure, which we record as a proposition.

Proposition 4.20 (The transmutation group and the universal transition operator)**.** *Identify each region with its predicate triple in* $\{0,1\}^3$ *(Theorem 4.8), and let* $\mathbb{Z}_2^3$ *act on the cube by coordinate-wise addition modulo* 2 *(predicate toggling),* $g \cdot t = g \oplus t$*; call* $\mathbb{Z}_2^3$ *under this action the* transmutation group*. Then:*

(i) *the action is* simply transitive *on the eight cells – for any ordered pair of cells* (u, v) *there is a unique* $g \in \mathbb{Z}_2^3$ *with* $g \cdot u = v$*, namely* $g = u \oplus v$ *– so between any two regions there is a unique* universal transition operator*, the group element toggling exactly the predicates on which they differ;*

(ii) *the three single-predicate toggles* $(1,0,0)$*,* $(0,1,0)$*,* $(0,0,1)$ *generate* $\mathbb{Z}_2^3$*, and the Cayley graph on these generators has diameter* 3*, equal to the rank* $\rho(R_7) = 3$ *of the full triple in Remark 4.19.*

Proof. (i) In the group $(\mathbb{Z}_2^3, \oplus)$ every element is its own inverse, so $g \cdot u = v$ reads $g \oplus u = v$, whose unique solution is $g = v \oplus u = u \oplus v$. Existence and uniqueness of g for each ordered pair (u, v) is exactly simple transitivity. (ii) Any $t = (t_1, t_2, t_3) \in \mathbb{Z}_2^3$ equals $t_1(1,0,0) \oplus t_2(0,1,0) \oplus t_3(0,0,1)$, so the three single-predicate toggles generate the group; the word length of t in these generators is its Hamming weight $|t| \leq 3$, attained by the all-ones element $(1,1,1)$, so the Cayley-graph diameter is 3. □

Figure 3 draws the restriction of this Cayley graph to the seven classified cells.

Remark 4.21 (Admissibility constraints and deferred realisation). Two constraints of the classification restrict which operators an *agent* may apply: the absolute exterior $(0,0,0)$ is excluded as a source cell (Remark 3.9), and operators toggling a cell into the suspended $R_3 = (0,0,1)$ are blocked until the K-objective reading reactivates them (Axiom 4.11; agent-relatively, Section 5). The universal transition operator of Proposition 4.20 is thus a purely *algebraic* object here. The present paper fixes *which* operators are admissible, and from and into which cells. The quantitative half of the question – how far a realisation carries, at what cost, and along what trajectory, together with the reachability and control of region transitions – is a dynamical question, and it is open: no cost model, trajectory, or control notion is supplied here or in the companion work, which treats restricted realization mechanisms of its own rather than movement along the transition operator [Bra26]. This section establishes only the static poset and symmetry structure.

5. Agents: Multi-Agent Relativization

The framework as developed so far characterizes a *single classifier*: one system assigns the predicate triple $(M(x), C(x), K(x))$ to each artifact $x \in \mathcal{A}$. This subsection relativizes the classification to a population of classifiers. It introduces *no new primitives and no new axioms*: an agent is an instantiation of the existing predicate triple, and every result below follows by parametric application of the definitions of Section 3 and Sections 4.1 to 4.6. Two phenomena emerge. First, *region non-invariance* (Lemma 5.20): the same artifact

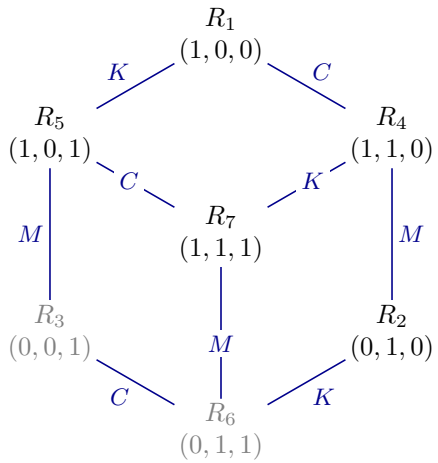


Figure 3: Adjacency of the seven classified cells: two regions are joined exactly when their triples differ in one coordinate, and the edge carries that predicate. Read towards the richer triple an edge adds the predicate, read back it removes it, so nine edges carry eighteen directed transitions. This is the Cayley graph of Proposition 4.20(ii) restricted to $\mathcal{A}^+$: the 3-cube has twelve edges, and deleting the absolute exterior $(0,0,0)$ with its three incident edges (Remark 3.9) leaves these nine – the drawing is the cube seen along its main diagonal, which is why no edge crosses another. Diameter 3 is realised on *opposite corners*: R_1/R_6, R_2/R_5, R_3/R_4, each a one-predicate cell and the two-predicate cell disjoint from it. Greyed: the suspended R_3 (Axiom 4.11) and R_6, its only K-neighbour outside M. *Edges are adjacencies of the classification, not operations performed on artifacts.*

may occupy different regions under different agents. Second, *conditional reactivation of* R_3 (Proposition 5.15), which discharges the forward pointer of Remark 4.13.

5.1. **Agent Model.**

Definition 5.1 (Semantic-functional agent)**.** An *agent* is a triple

$$a = (M_a, C_a, K_a), \qquad M_a, C_a, K_a \subseteq \mathcal{A},$$

where M_a, C_a, K_a are the sets of artifacts for which the respective predicates (Definitions 3.2 to 3.4) hold under agent a's classification. That is, for each $x \in \mathcal{A}$ (Definition 3.1),

$$M_a(x) = [x \in M_a], \quad C_a(x) = [x \in C_a], \quad K_a(x) = [x \in K_a],$$

using Iverson-bracket notation.

Remark 5.2 (What "agent" does and does not import). The word names the triple and nothing further. The framework imposes on whatever bears such a triple no condition of agency, autonomy, intentionality, symbol-possession or functional capacity, and it takes no position on which triples describe an agent in any substantive sense: Definition 5.1 is a naming convention for an instantiation of the predicates of Section 3, adopted because a population of such instantiations (Definition 5.4) is the intended application. Where a statement below says what an agent has, lacks, or cannot do – Definition 6.4, Corollary 7.5, and Theorem 8.22 – the statement is about the extensions M_a, C_a, K_a and which cells they

inhabit, not about a subject; the passages that instead speak of an agent acting, behaving, or representing itself are conjectural, and no result rests on them.

Remark 5.3 (Minimality: specialization, not redefinition). The single-classifier framework of Sections 3 and 4 is recovered by fixing one agent a and writing $M = M_a$, $C = C_a$, $K = K_a$: the agent layer *specializes* the existing definition points rather than redefining them.

Definition 5.4 (Agent population)**.** An *agent population* is a finite indexed family

$$\mathfrak{A} = \{a_1, \ldots, a_n\}$$

of agents (Definition 5.1) over a shared artifact domain $\mathcal{A}$ (Definition 3.1).

Remark 5.5 (Shared domain, private predicates). All agents classify the same artifact domain $\mathcal{A}$. The predicates M_a, C_a, K_a are agent-private: they record the semantic-functional state of agent a with respect to each artifact. No assumption is made about agreement between agents.

5.2. **Relativized Regions.**

Definition 5.6 (Relativized regions and Θ_a)**.** For an agent a (Definition 5.1) and each $i \in \{1, \ldots, 7\}$, the *relativized region* R_i^a is the region R_i of the framework (Definitions 4.1 to 4.7) instantiated on the predicate triple (M_a, C_a, K_a). The *relativized classified domain* is

$$\mathcal{A}_a^+ = \{x \in \mathcal{A} \mid M_a(x) \vee C_a(x) \vee K_a(x)\}$$

(the per-agent instantiation of Definition 3.7), and for $x \in \mathcal{A}_a^+$ the *region assignment* $\mathrm{Region}_a(x)$ is the unique R_i^a determined by the triple $(M_a(x), C_a(x), K_a(x))$ according to the standard region correspondence. The triple $(0, 0, 0)$ under a is not a region: it is the *absolute exterior under* a, written $\mathrm{Region}_a(x) = \bot$. Finally, the *relativized structured ambiguity domain* is the per-agent instantiation of Definition 4.16:

$$\Theta_a = M_a \cap C_a = R_4^a \cup R_7^a.$$

Proposition 5.7 (Relativized partition)**.** *For each agent a, the relativized regions $R_1^a, \ldots, R_7^a$ partition the relativized classified domain $\mathcal{A}_a^+$.*

Proof. Identical to Theorem 4.8, applied to the predicate triple (M_a, C_a, K_a) in place of (M, C, K). □

Definition 5.8 (Derived agent sets)**.** For each agent a (Definition 5.1):

$$\mathrm{KNOWN}_a = M_a \cup C_a, \qquad \mathrm{UNKNOWN}_a = \{x \in \mathcal{A} \mid M_a(x) = 0 \wedge C_a(x) = 0\}.$$

These are the per-agent instantiations of Definition 3.5 and inherit all properties established in Section 3.3: in particular, KNOWN_a and $\mathrm{UNKNOWN}_a$ partition $\mathcal{A}$ (Proposition 3.6), and $\mathrm{UNKNOWN}_a$ stratifies as in Proposition 3.8, both applied to (M_a, C_a, K_a).

5.3. **Relative Knowledge.**

Definition 5.9 (Regional disagreement / relative knowledge)**.** For agents a, b (Definition 5.1) and an artifact $x \in \mathcal{A}_a^+ \cup \mathcal{A}_b^+$ (Definition 5.6), a and b *disagree on* x if

$$\mathrm{Region}_a(x) \neq \mathrm{Region}_b(x).$$

This includes the case where $x \in \mathcal{A}_a^+$ but $x \notin \mathcal{A}_b^+$ (or vice versa), i.e., one agent classifies x while the other places it in its absolute exterior ($\mathrm{Region}_b(x) = \bot$). *Alias:* the *relative knowledge property* $\mathrm{KR}(x, a, b)$ is this same predicate, $\mathrm{KR}(x, a, b) \overset{\text{def}}{\iff} \mathrm{Region}_a(x) \neq \mathrm{Region}_b(x)$; the two names denote one relation (disagreement stated as a property of the artifact–agent pair).

Remark 5.10 (Interpretation of KR). KR captures the fact that regional classification is *agent-dependent*. The "knowledge" in "relative knowledge" is not an additional predicate: it is the observation that the existing MCK classification, when parametrized by agents, produces different region assignments for the same artifact. This is a structural consequence of predicate privacy (Remark 5.5), not a new semantic primitive.

Remark 5.11 (Boundary conditions). (i) *Trivial agreement.* If $a = b$ (identical predicate triples), then $\mathrm{KR}(x, a, b)$ is false for all x. (ii) *Total disagreement.* If $M_a = C_b$, $C_a = M_b$, $K_a = K_b$ (agents with swapped semantic integration), then $\mathrm{KR}(x, a, b)$ is true for all $x \in M_a \triangle M_b$ (symmetric difference). (iii) *Single-predicate divergence.* If the agents agree on two predicates for x but differ on the third, then $\mathrm{Region}_a(x) \neq \mathrm{Region}_b(x)$ (Lemma 5.24).

5.4. **Functional Potential and the Two Readings of K.**

In the remainder of this subsection, a and b denote agents (Definition 5.1) and f denotes an artifact; a is the agent whose horizon f lies outside, and b is the witness.

Definition 5.12 (Functional potential)**.** An artifact f exhibits *functional potential* relative to agent a and witness agent b (Definition 5.1) if

$$\mathrm{PF}(f, a, b) \iff f \in R_5^b \ \wedge \ f \in R_3^a,$$

with the relativized regions as in Definition 5.6 and the conjunct $f \in R_3^a$ read under the K-objective interpretation (Definition 5.14). Explicitly: agent b classifies f as a transparent operator ($M_b(f) = 1$, $C_b(f) = 0$, $K_b(f) = 1$): b both understands and can execute f. Agent a classifies f with $(M_a(f), C_a(f), K_a(f)) = (0, 0, 1)$: f is functional in $\mathcal{A}$ but entirely outside a's semantic horizon – a neither understands f nor even recognizes it, yet f is genuinely functional, as witnessed by b.

Remark 5.13 (Why R_3^a is well-defined here). In the single-agent framework, R_3 is suspended (Axiom 4.11) because the conjunction $K(f) = 1 \wedge M(f) = 0 \wedge C(f) = 0$ is self-referentially problematic: how can the single classifier assert $K(f) = 1$ for an artifact it does not semantically recognize? In the multi-agent setting this problem dissolves: the assertion $K_a(f) = 1$ is not made by agent a – it records that f is functional with respect to the operational infrastructure accessible to a, even though a has no semantic access to f. Making this precise requires distinguishing two interpretations of K, which is the content of Definition 5.14. The distinction is itself a formal contribution: it disambiguates the R_3 suspension and provides the conditions under which R_3 can be reactivated.

Definition 5.14 (K-objective vs. K-subjective)**.** For an agent a (Definition 5.1) and an artifact f:

- $K_a^{\text{subj}}(f) = 1$ iff a can execute f *and* a recognizes f as functional. This implies $M_a(f) = 1$ or $C_a(f) = 1$.
- $K_a^{\text{obj}}(f) = 1$ iff f is executable on some architecture within a's operational scope, *regardless* of a's semantic access.

Under the K-subjective reading, $R_3^a = \emptyset$ for every agent a: the implication $M_a(f) = 1 \vee C_a(f) = 1$ excludes the $(0, 0, 1)$ triple, reproducing the suspension of Axiom 4.11 as a per-agent consequence of the definition. Under the K-objective reading, R_3^a may be non-empty, and functional potential (Definition 5.12) becomes definable.

Proposition 5.15 (R_3 reactivation criterion)**.** *Under the K-objective reading of Definition 5.14, $R_3^a \neq \emptyset$ if and only if there exists $f \in \mathcal{A}$ such that: (i) f is executable on an architecture accessible to a, and (ii) $M_a(f) = 0$ and $C_a(f) = 0$.*

Proof. Direct from Definition 4.3 applied to the triple $(M_a, C_a, K_a^{\text{obj}})$. Condition (i) is $K_a^{\text{obj}}(f) = 1$ by Definition 5.14; together with (ii) it places f in the $(0, 0, 1)$ cell, i.e., $f \in R_3^a$. Conversely, any $f \in R_3^a$ satisfies $K_a^{\text{obj}}(f) = 1$ and $M_a(f) = C_a(f) = 0$, which are exactly (i) and (ii). □

Remark 5.16 (The un-elicited reservoir: R_3^a and its static counterpart). We conjecture that Proposition 5.15 locates R_3^a within a broader phenomenon. Relative to an agent a, call the *un-elicited reservoir* the content that a has not yet been queried, as an observer, to surface – everything presently outside a's semantic body ($M_a = C_a = 0$, Definition 5.8) that a demand on a could actualise. The region R_3^a is the *purely functional* slice of this reservoir: latent capability (the K-objective reading), awaiting elicitation. But we conjecture the reservoir is not only functional. It equally holds latent *static* content – data that, once elicited, resolves as comprehensible (into M_a), ambiguous (into $\Theta = M_a \cap C_a$), or opaque/obfuscated (into C_a), alongside content a query leaves un-understood. Which region a latent element enters is fixed by its content, not by the act of elicitation; functionality (R_3) is thus one stratum of the un-queried reservoir, not the whole of it – the reservoir spans the full unknown stratified in Proposition 3.8. The external witness b of Definition 5.12 is what does so in that definition. The elicitation is a dynamical, not a static, act, and its region-transition content is developed neither here nor in the companion on region transitions [Bra26], which treats restricted realization mechanisms of its own; it is open. One clause is settled later: Theorem 8.22(iii) proves that nothing outside a's semantic body can be the subject of a's own queries, so the reservoir is un-elicited in exactly that sense. The latent-static-content clause is untouched and stands as conjectured.

Remark 5.17 (Compatibility with the single-agent framework). The single-agent framework uses K-subjective implicitly: Remark 4.12 argues that functionality without semantic presence is problematic. The multi-agent extension does not override this; it *adds* the K-objective interpretation as a second reading, applicable when external witnesses exist. The existing theory under K-subjective remains valid as a special case.

Remark 5.18 (Operational reading: function without key). In operational terms, K_a^{obj} separates the transformation itself (the *function*) from the agent-specific semantic access required to recognize it (the *key*): $f \in R_5^b$ means agent b holds both function and key; $f \in R_3^a$ means

agent a holds the (objective) function but no key. The existence of an execution of f within a's operational scope is independent of whether a possesses the key.

Remark 5.19 (Per-agent recovery of the suspended cell). Definition 5.14 is the *per-agent recovery of the excluded configuration* flagged in Remark 4.14: the $(0,0,1)$ predicate cell, excluded from operational instantiation by Axiom 4.11, is here recovered per agent. This discharges the forward pointer of Remark 4.13. Under the K-subjective reading, the emptiness of R_3^a is now a provable statement rather than a convention (Definition 5.14); under the K-objective reading, R_3^a can be non-empty (Proposition 5.15), and its inhabitants are exactly the functional-potential artifacts of Definition 5.12. The excluded cell is thereby acknowledged in both readings; no silent exclusion remains.

5.5. **Lemmas.**

Lemma 5.20 (Region non-invariance)**.** *For any artifact domain $\mathcal{A}$ with $|\mathcal{A}| \geq 1$, there exist agents a, b (Definition 5.1) and an artifact x such that*

$$\mathrm{Region}_a(x) \neq \mathrm{Region}_b(x).$$

Proof (constructive witness). Let $x \in \mathcal{A}$ be any artifact. Define agents

$$a = (\{x\}, \emptyset, \emptyset), \qquad b = (\emptyset, \{x\}, \emptyset).$$

Then $\mathrm{Region}_a(x) = R_1^a$ (triple $(1,0,0)$) and $\mathrm{Region}_b(x) = R_2^b$ (triple $(0,1,0)$). Since $R_1 \neq R_2$, we have $\mathrm{Region}_a(x) \neq \mathrm{Region}_b(x)$, i.e., $\mathrm{KR}(x, a, b)$ holds (Definition 5.9). □

In the witness, the same artifact x receives triple $(1,0,0)$ under $a = (\{x\}, \emptyset, \emptyset)$ and $(0,1,0)$ under $b = (\emptyset, \{x\}, \emptyset)$, so $\mathrm{Region}_a(x) = R_1^a \neq R_2^b = \mathrm{Region}_b(x)$: region assignment is a relation over $\mathcal{A} \times \mathfrak{A}$, not a property of the artifact alone (Remark 5.21). The witness is deliberately minimal; Example 5.29 exhibits the same phenomenon on a non-degenerate artifact.

Remark 5.21 (Non-triviality and consequences). The same artifact – identical in $\mathcal{A}$ – is pure understanding for one agent and pure opacity for another: agent a has semantically integrated x, while agent b recognizes x but has no structured access to it. Although the construction is elementary, the result is not trivial: it establishes that region classification is a *relation*

$$\mathcal{A} \times \mathfrak{A} \to \{R_1, \ldots, R_7, \bot\},$$

not a property of artifacts alone. Consequently, any downstream use of MCK regions must treat claims about "the region of x" as meaningful only relative to a specified agent.

Lemma 5.22 (R_3 emergence via multi-agent relativization)**.** *There exist agents a, b and an artifact f such that*

$$f \in R_5^b \ \wedge \ f \in R_3^a$$

under the K-objective interpretation of Definition 5.14; equivalently, the predicate $\mathrm{PF}(f, a, b)$ *of Definition 5.12 is satisfiable.*

Witness construction. Let $f \in \mathcal{A}$ be a functional artifact (e.g., a sorting algorithm). Define:

- Agent $b = (M_b, C_b, K_b)$ with $M_b = \{f\}$, $C_b = \emptyset$, $K_b = \{f\}$. Triple for f: $(1, 0, 1)$, hence $f \in R_5^b$: agent b both executes and understands f.
- Agent $a = (M_a, C_a, K_a^{\text{obj}})$ with $M_a = \emptyset$, $C_a = \emptyset$, $K_a^{\text{obj}} = \{f\}$. Triple for f: $(0, 0, 1)$, hence $f \in R_3^a$.

The objective functionality is justified: f is executable on architectures within a's scope (the sorting algorithm runs on a's hardware), but a has no semantic access – a does not know the algorithm exists, cannot name it, and has never encountered it.

Remark 5.23 (Epistemic boundary of Lemma 5.22). This instance is witnessed, not proven, and is not to be inflated: the construction formally exhibits the predicate assignment, but the premise $K_a^{\text{obj}}(f) = 1$ rests on an *architectural* fact – executability of f within a's operational scope – that is external to the formal system of Sections 3 and 4. The witness certifies coherence of the K-objective reading in a concrete instance; it does not derive that reading from the axioms, and a uniform proof from the axioms is left to future work. The interpretation is that R_3 is the region of *functional potential*: concrete operational capacity that exists in the shared artifact domain but has not entered a particular agent's semantic horizon. The external witness b certifies the functionality, resolving the semantic circularity that motivated the single-agent suspension.

Lemma 5.24 (Single-predicate divergence)**.** *If agents a, b agree on two predicates for an artifact x but disagree on the third, then* $\text{Region}_a(x) \neq \text{Region}_b(x)$.

Proof. The region map $\rho\colon \{0,1\}^3 \setminus \{(0,0,0)\} \to \{R_1, \ldots, R_7\}$ is a bijection (Theorem 4.8). The two agents' triples for x differ in exactly one coordinate, hence are distinct elements of $\{0,1\}^3$. If both triples are non-zero, they map under ρ to distinct regions. If one of them is $(0,0,0)$, the corresponding agent places x in its absolute exterior ($\text{Region}(x) = \perp$), which differs from every region; this counts as disagreement per Definition 5.9. In either case $\text{Region}_a(x) \neq \text{Region}_b(x)$. □

Corollary 5.25 (Bound on distinct assignments)**.** *In a population of n agents (Definition 5.4), the number of distinct region assignments in $\{R_1, \ldots, R_7\}$ received by a single artifact x is at most* $\min(n, 7)$ *– bounded by the number of non-zero triples in $\{0,1\}^3$. The bound is tight in a reading-dependent way: under the K-objective reading (Definition 5.14) the value* $\min(n, 7)$ *is attained; under the K-subjective reading the attainable maximum is* $\min(n, 6)$.

Proof. Each agent assigns x at most one region (Proposition 5.7), and there are only seven regions (Theorem 4.8), giving the upper bound $\min(n, 7)$. For attainability: under the K-objective reading the value $\min(n, 7)$ is realized by choosing agents with pairwise distinct non-zero triples for x; under the K-subjective reading $R_3^a = \emptyset$ for every agent (Definition 5.14), so the $(0, 0, 1)$ cell is unrealizable, only six triples remain available, and the attainable maximum is $\min(n, 6)$. □

5.6. **Semantic Absence and Collapse.**

Definition 5.26 (Semantic absence)**.** For an agent a (Definition 5.1) and an artifact $x \in \mathcal{A}$:

$$U_a(x) \iff \neg M_a(x) \wedge \neg C_a(x).$$

Equivalently, $U_a = \text{UNKNOWN}_a$ (Definition 5.8), now explicitly indexed by the agent: it is the complement of agent a's semantic body $\text{KNOWN}_a = M_a \cup C_a$ – what lies *outside* that body, neither integrated (M_a) nor apprehended (C_a). Writing $X^a_{\text{out}} = \mathcal{A} \setminus \mathcal{A}^+_a$ for the absolute exterior under a (the per-agent instantiation of the exterior named in Definition 3.7 and Remark 3.9), the stratification Proposition 3.8, applied to (M_a, C_a, K_a), yields the disjoint decomposition

$$U_a = R^a_3 \cup X^a_{\text{out}}.$$

U_a is a *derived set*, not a new region: it is the agent-relative projection of the existing UNKNOWN partition onto agent a's predicate assignment. It introduces no new ontological commitment and does not alter the seven-region partition (Theorem 4.8), which applies only to $\mathcal{A}^+_a$.

Lemma 5.27 (Semantic collapse)**.** *For every agent a and artifact x:*

$$x \in R^a_3 \;\Rightarrow\; U_a(x), \qquad x \in X^a_{\text{out}} \;\Rightarrow\; U_a(x),$$

and the two components are disjoint and exhaust U_a.

Proof. Both $R^a_3 = \{x \mid M_a(x) = 0,\ C_a(x) = 0,\ K_a(x) = 1\}$ and $X^a_{\text{out}} = \{x \mid M_a(x) = 0,\ C_a(x) = 0,\ K_a(x) = 0\}$ satisfy $\neg M_a(x) \wedge \neg C_a(x)$, which is exactly $U_a(x)$ (Definition 5.26). Disjointness and exhaustiveness follow from Proposition 3.8 applied to (M_a, C_a, K_a): within U_a, either $K_a(x) = 1$ (placing x in R^a_3) or $K_a(x) = 0$ (placing x in X^a_{out}). □

Remark 5.28 (Indistinguishability from the agent's perspective). For any $x \in U_a$ the two components of the decomposition differ only in the value of $K_a(x)$ – a value agent a cannot assess subjectively: by Definition 5.14, $K^{\text{subj}}_a(x) = 1$ would imply $M_a(x) = 1$ or $C_a(x) = 1$, contradicting $x \in U_a$. Hence, within a's own semantic state, R^a_3 and X^a_{out} are indistinguishable: the collapse is complete. Separating them is therefore a question the subjective reading does not decide; how the K-objective value is obtained is not settled here. Lemma 5.22 exhibits one configuration in which it is available – an agent b with $f \in R^b_5$ – which shows the objective reading can differ, not that a second agent is the only route to it.

Example 5.29 (The same function in different regions). Let f be a standard encryption function. For a cryptographer b: $M_b(f) = 1$ (understands the algorithm), $C_b(f) = 0$, $K_b(f) = 1$ (can execute it), so $f \in R^b_5$. For an end user a: $M_a(f) = 0$, $C_a(f) = 1$ (recognizes "encryption" as a concept), $K_a(f) = 1$ (can invoke it as an API call), so $f \in R^a_6$. Here $\text{KR}(f, b, a)$ holds (Definition 5.9): the same function is a transparent operator for the cryptographer and an opaque operator for the end user – an instance of Lemma 5.20 on the binary regions.

5.7. **Agent-Relative Denotation.**

Definition 5.30 (Agent-relative denotation)**.** For each agent a (Definition 5.1), the *agent-relative denotation function*

$$\text{denotes}_a \colon \Theta_a \to \text{Ref}$$

is the instantiation of Definition 4.17 on a's predicate triple: it maps each artifact $s \in \Theta_a$ (Definition 5.6) – a *symbol for a* – to a referent, every denotation target lying in a's classified

domain $\mathcal{A}_a^+$ within a's active regions. It inherits from Definition 4.17 the three properties: (i) *static*; (ii) *single-valued*, the full set of targets of a symbol being collected by the per-agent instantiation of Remark 4.18; and (iii) *domain-restricted* – only Θ_a-artifacts denote for a: what counts as a symbol is itself agent-relative. No cross-agent coherence is assumed: for $s \in \Theta_a \cap \Theta_b$, the referents $\text{denotes}_a(s)$ and $\text{denotes}_b(s)$ may differ – denotation, like region assignment (Lemma 5.20), is agent-relative. The single-classifier function of Definition 4.17 is recovered by fixing one agent.

Remark 5.31 (Invariance of the poset, variance of the assignment). Definition 5.30 is stated explicitly so that downstream work referring to per-agent symbol reference has a single definition point; like every result of this section, it introduces no new primitive. The region poset of Section 4.6 is invariant across agents – the abstract lattice $2^{\{M,C,K\}} \setminus \{\emptyset\}$ does not change – while the assignment $x \mapsto R_i^a$ varies: the poset is a property of the framework, the assignment a property of the agent–artifact pair.

6. The Semantic Horizon and Capability Profile

The definitional core is now in place. This section names the boundary the preceding sections have been implicitly using and reads an agent's region inventory as its capability profile; Section 7 then classifies the qualitative types that profile can take, Section 8 fixes which movements between cells an agent can coherently realize, and Sections 9 and 10 position the framework and record the limitations and open questions.

6.1. The Semantic Horizon.

The partition of the artifact domain into KNOWN and UNKNOWN (Definition 3.5 and Proposition 3.6) induces a boundary to which earlier sections have already referred informally – the point at which an artifact "enters the semantic horizon" (Remark 4.12) or lies "outside x's semantic horizon" (Definition 5.12). This section names that boundary and consolidates in one place the facts already established about it. It is a synthesis over Sections 3.3 and 5: no new machinery, no new axioms, and – beyond the naming definition below – no new claims.

Definition 6.1 (Semantic horizon)**.** The *semantic horizon* is the boundary of the known domain: the separation of KNOWN $= M \cup C$ from UNKNOWN established by Definition 3.5. An artifact $x \in \mathcal{A}$ (Definition 3.1) is *inside* the horizon iff $M(x) = 1 \vee C(x) = 1$, and *outside* the horizon iff $M(x) = 0 \wedge C(x) = 0$. This definition *names* an existing derived boundary; it introduces no new predicate and no set beyond those of Definition 3.5. In particular, horizon membership is decided by M and C alone: K has functional, not semantic, character (Definition 3.4), and $K(x) = 1$ by itself does not place x inside the horizon.

Crossing into classification. An artifact is inside the horizon exactly when at least one of the two semantic predicates fires: recognition (C) or semantic integration (M). Every artifact inside the horizon is assigned exactly one region (Corollary 4.10); the horizon is thus also the boundary of guaranteed regional classifiability for semantically present artifacts.

Stratification of the outside. The outside of the horizon is not homogeneous. By Proposition 3.8,

$$\text{UNKNOWN} = R_3 \cup (\mathcal{A} \setminus \mathcal{A}^+),$$

a disjoint decomposition into two strata that must not be conflated:

- the *suspended pure-K cell* R_3 (Definition 4.3) – artifacts functionally active but semantically absent, formally defined yet operationally suspended (Axiom 4.11), an explicitly acknowledged boundary case (Remark 4.14);
- the *absolute exterior* $\mathcal{A} \setminus \mathcal{A}^+$ (Definition 3.7) – artifacts for which no predicate holds at all, likewise acknowledged explicitly (Remark 3.9).

Both strata lie outside the horizon, but for different reasons: the first is invisible to the classifier while remaining functionally present in the domain; the second is entirely beyond the framework's classification.

Agent-relative horizon. Under the multi-agent relativization of Section 5, the horizon relativizes with everything else: each agent a has its own horizon, the boundary of KNOWN_a (Definition 5.8), whose outside is the semantic absence U_a (Definition 5.26) with its per-agent stratification $U_a = R_3^a \cup X_{\text{out}}^a$ (Lemma 5.27). From inside agent a's own semantic state the two strata are indistinguishable – the semantic collapse of Remark 5.28; the objective reading may separate them (Lemma 5.22), and this section does not treat how that value is obtained.

Remark 6.2 (Horizon non-invariance). The same artifact may be inside one agent's horizon and outside another's. Witness: for any $x \in \mathcal{A}$, take $a = (\{x\}, \emptyset, \emptyset)$ and $b = (\emptyset, \emptyset, \emptyset)$ (Definition 5.1); then $x \in \text{KNOWN}_a$ while $U_b(x)$ holds (Definitions 5.8 and 5.26). This is the horizon-level instance of region non-invariance (Lemma 5.20): like the region assignment, horizon membership is a property of the agent–artifact pair, not of the artifact alone.

Remark 6.3 (Boundary: horizon dynamics deferred). This section is static, consistent with the scope conventions of Section 3: it names the horizon at a fixed semantic state. How artifacts *cross* the horizon over time – promotion into recognition when C first fires, into integration when M fires, and the corresponding entry into the region system – is a region-transition topic, deferred to the same downstream treatment as the region dynamics (Remark 4.21). No transition formalism is introduced here.

6.2. **Capability Profile.** The agent-relative layer establishes that region assignment is a relation over artifact and agent (Lemma 5.20), not a property of the artifact alone. The operational consequence the proven machinery already earns is this: an agent's inventory of regions *is* its capability profile, and that inventory can fail in two distinct ways – a region can be absent, or it can be inhabited under one reading of K and left undecided by the other. No new primitive and no new proof are introduced; Definition 6.4 and Corollary 6.6 *consume* results already established in Section 5.

Definition 6.4 (Capability profile)**.** For an agent a (Definition 5.1) over its artifact domain, the *capability profile* of a is the set of MCK regions inhabited for a:

$$\text{Cap}(a) = \{\, i \in \{1, \ldots, 7\} \mid R_i^a \neq \emptyset \,\},$$

where R_i^a is the relativized region of Definition 5.6. A region $i \notin \text{Cap}(a)$ is said to be *absent for* a.

Remark 6.5 (A missing region is a limitation of reach). Unfolding the definition gives the capability reading directly: $i \notin \mathrm{Cap}(a)$ means $R_i^a = \emptyset$, so no artifact carries, under a's predicate triple (M_a, C_a, K_a), the combination defining R_i (Definitions 4.1 to 4.7) – and that combination is exactly what would license a to classify an artifact into R_i and, through the K-component, to act on it. A missing region is therefore a limitation of a's classificatory and operational reach. Because region assignment is agent-relative and non-invariant (Lemma 5.20), the absence is a property of a, not of the artifact domain: non-invariance supplies a witness in which the very same artifacts populate R_i^b for another agent b.

Corollary 6.6 (The subjective reading does not decide R_3^a)**.** *Under the K-subjective reading of Definition 5.14, agent a's own predicate triple does not separate R_3^a – functional for a, yet neither meaningful nor recognized as opaque by a – from the absolute exterior X_{out}^a: within U_a the two components differ only in a value that reading does not supply. Hence whether $R_3^a \neq \emptyset$ under the K-objective reading is not a function of $(M_a, C_a, K_a^{\mathrm{subj}})$: where $U_a \neq \emptyset$, any determination of it appeals to information outside that triple. (Where $U_a = \emptyset$ the question is settled by M_a and C_a alone, since $R_3^a \subseteq U_a$.)*

Proof. Immediate from Lemma 5.27 and Remark 5.28. Within $U_a = R_3^a \cup X_{\mathrm{out}}^a$ the two components differ only in the value of $K_a(x)$; under the K-subjective reading $K_a^{\mathrm{subj}}(x) = 1$ would force $M_a(x) = 1$ or $C_a(x) = 1$ (Definition 5.14), contradicting $x \in U_a$. That reading therefore assigns 0 throughout U_a and separates nothing in it. The K-objective value is a distinct assignment; how it is obtained is not at issue here. □

Scope. Corollary 6.6 concerns one reading of K and settles one question about it: the subjective reading leaves R_3^a undecided. It says nothing about how the K-objective value is obtained (Remark 5.23) – and in particular nothing about what an agent can or cannot establish about its own configuration, a question this paper does not treat, on the boundary stated in Section 7.2 and recorded as a limitation in Section 10.

Remark 6.7 (Epistemogenic self-representation). We conjecture that an agent that represents its own MCK configuration – that forms an artifact standing for "my own (M, C, K) assignment" – thereby brings that self-artifact into $M \cap C \cap K = R_7$ of itself: the self-model is meaning-bearing (M), retains unresolved opacity about its own hidden structure (C), and is operative on the agent's behaviour (K). On this conjecture, self-representation is *epistemogenic*: it *creates* the full-triple region of the self rather than merely reporting it. This is offered as a conjecture; it introduces no primitive and is used by no result above. A formal treatment of the reflexive act and its cost is left to downstream work.

7. Agentic Configurations

The agent model of Section 5 equips each agent a with a predicate triple (M_a, C_a, K_a), and hence with three *extensions* – the subsets $M_a, C_a, K_a \subseteq \mathcal{A}$ of the artifact domain on which each predicate holds (Definition 5.1; on what the word does not import, Remark 5.2). Because these are three arbitrary sets, an agent's semantic–functional state has a coarsest invariant: *which* of the seven relativized regions of Definition 5.6 are inhabited. This section classifies those invariants (Section 7.1), draws the consequences for an agent that the framework already licenses – the placement of symbols and the status of the generative unknown

under the two readings of K – together with one conjecture about the reflexive act of self-representation (Section 7.2), and isolates the coarser datum that the type alone determines (Section 7.3).

7.1. **The Qualitative Types of an Agent.**

Definition 7.1 (Qualitative type)**.** The *qualitative type* $\tau(a)$ of an agent a (Definition 5.1) *is* the capability profile of Definition 6.4, the set of indices of its non-empty relativized regions,

$$\tau(a) \;=\; \mathrm{Cap}(a) \;=\; \{\, i \in \{1,\dots,7\} \mid R_i^a \neq \emptyset \,\},$$

with R_i^a as in Definition 5.6. This is one object under two readings, not two objects: Definition 6.4 reads the set as an *inventory* – which regions the agent has available, and which are absent to it – while this section reads the very same set as a combinatorial *type*, the datum by which agents are classified and counted. The notation $\tau(a)$ marks the second reading and introduces no new primitive. Two agents have the same type when the same cells of the partition (Proposition 5.7) are inhabited for both, irrespective of *which* artifacts inhabit them.

The type is exactly the datum that Theorem 4.8 licenses one to read off an agent: the partition assigns each classified artifact to one cell, so "which cells are occupied" is well defined and is the only agent-level invariant that the static classification supplies. The following theorem determines the types completely.

Theorem 7.2 (Classification of qualitative types)**.** *Read the agent's triple K-objectively (Definition 5.14), so that R_3^a may be inhabited; the counts below are counts under that reading. Let $S \subseteq \{1,\dots,7\}$. There exists an agent a with M_a, C_a, K_a all non-empty and $\tau(a) = S$ (Definition 7.1) if and only if*

$$S \cap \{1,4,5,7\} \neq \emptyset, \qquad S \cap \{2,4,6,7\} \neq \emptyset, \qquad S \cap \{3,5,6,7\} \neq \emptyset,$$

i.e. S meets the cells on which M, on which C, and on which K is active. Exactly 109 *of the* $2^7 = 128$ *subsets of* $\{1,\dots,7\}$ *satisfy this; up to permutation of the three predicates, there are exactly* 34 *types.*

Remark 7.3 (The admissible types are the covers of the predicate set). The classification has a shorter description, and it is worth giving because it locates the counts in a classical family rather than presenting them as new. Identify each region with the non-empty subset of $\{M, C, K\}$ that holds on it, so that $R_1 \mapsto \{M\}$, $R_4 \mapsto \{M, C\}$, $R_7 \mapsto \{M, C, K\}$, and so on. A type is then a set of non-empty subsets of a three-element set, and the three admissibility conditions say exactly that the union of those subsets is all of $\{M, C, K\}$: the extensions M_a, C_a, K_a are non-empty precisely when each of the three predicates is carried by some inhabited cell. *The admissible types are therefore exactly the covers of* $\{M, C, K\}$, and 109 and 34 are the labelled and unlabelled cover numbers of a three-element set [Com74, BLL98]. The proof above is the standard derivation of those numbers – inclusion–exclusion for the first, Burnside for the second – specialised to $n = 3$. What this paper contributes at this point is the identification, not the arithmetic: an agent's semantic–functional state is a cover of its own predicate set, and the results below are about which covers do what.

Two counts are in play here and should not be confused. A single artifact carries one predicate triple, so it falls in one of $2^3 = 8$ cells, of which the seven regions are those remaining once the absolute exterior $(0, 0, 0)$ is set aside (Theorem 4.8): that bound governs *artifacts.*

An agent, by contrast, is characterised by *which* of those seven regions are inhabited for it – one binary datum per region, hence 2^7 patterns a priori. The theorem says that 19 of those patterns are unrealizable, each because it would empty one of the three extensions. The non-emptiness hypothesis is a modelling choice rather than a bookkeeping device, and this is the place to say why it is the right one. An entity one of whose three extensions is wholly empty is not exercising that axis at all – nothing it holds is understood ($M_a = \emptyset$), or nothing it holds is left unresolved ($C_a = \emptyset$), or nothing it holds is operative ($K_a = \emptyset$) – so its type reports nothing about that axis, and a classification of a classifier says something about all three only once each of them is inhabited at least once. That is the weakest such condition, and the 19 excluded patterns are exactly those that fail it: a subject holding only data, with no functionality, is the natural instance, since $K_a = \emptyset$ forces $\tau(a) \subseteq \{1, 2, 4\}$ and all eight such patterns are excluded. The reading of K bears on the total, which is why the statement declares one: the sufficiency construction below places an artifact in R_3^a whenever $3 \in S$, which the K-objective reading permits and the K-subjective reading does not (Definition 5.14). Under the subjective reading only the types with $3 \notin S$ survive, 52 of them, a restriction taken up in Section 7.2 and again in Section 8.

Proof. Necessity. By Theorem 4.8 applied to (M_a, C_a, K_a), $M_a = R_1^a \cup R_4^a \cup R_5^a \cup R_7^a$, so $M_a \neq \emptyset$ forces S to meet $\{1, 4, 5, 7\}$; likewise $C_a = R_2^a \cup R_4^a \cup R_6^a \cup R_7^a$ and $K_a = R_3^a \cup R_5^a \cup R_6^a \cup R_7^a$.

Sufficiency. Given such an S, take $\mathcal{A}$ to contain one artifact x_i for each $i \in S$, and let M_a (resp. C_a, K_a) consist of those x_i whose defining predicate triple (Definitions 4.1 to 4.7) has first (resp. second, third) coordinate 1. Each x_i then carries exactly the triple of R_i, so $R_i^a = \{x_i\}$ for $i \in S$ and $R_i^a = \emptyset$ otherwise, and the three conditions make M_a, C_a, K_a non-empty.

Count. Write A, B, C for the failures of the three conditions. A holds iff $S \subseteq \{2, 3, 6\}$, B iff $S \subseteq \{1, 3, 5\}$, C iff $S \subseteq \{1, 2, 4\}$, giving $|A| = |B| = |C| = 2^3 = 8$; the pairwise intersections are $S \subseteq \{3\}$, $S \subseteq \{2\}$, $S \subseteq \{1\}$, of size 2 each, and $A \cap B \cap C$ is $\{\emptyset\}$, of size 1. By inclusion–exclusion the number of excluded subsets is $24 - 6 + 1 = 19$, leaving $128 - 19 = 109$.

Up to relabelling. The symmetric group S_3 permutes the three predicates and hence acts on the seven cells by permuting the coordinates of their triples; the three conditions above are preserved, so the action restricts to the 109 admissible sets. The identity fixes all 109. A transposition, say of C and K, fixes R_1, R_6, R_7 and swaps $R_2 \leftrightarrow R_3$ and $R_4 \leftrightarrow R_5$, so an invariant S is a union of the five orbits $\{1\}, \{2, 3\}, \{4, 5\}, \{6\}, \{7\}$: of these $2^5 = 32$ sets, those failing the M-condition are the 4 avoiding $\{1\}, \{4, 5\}, \{7\}$, those failing the C-condition (equivalently, on invariant sets, the K-condition) are the 2 avoiding $\{2, 3\}, \{4, 5\}, \{6\}, \{7\}$, and 1 fails both; $32 - (4 + 2 - 1) = 27$. A three-cycle has orbits $\{1, 2, 3\}, \{4, 5, 6\}, \{7\}$, so an invariant S is one of $2^3 = 8$ unions, of which only $S = \emptyset$ is inadmissible: 7. By Burnside's lemma the number of orbits is $\frac{1}{6}(109 + 3 \cdot 27 + 2 \cdot 7) = \frac{204}{6} = 34$. □

Corollary 7.4 (Extremes of the classification)**.** *Among agents with M_a, C_a, K_a all non-empty, as in Theorem 7.2, the number $|\tau(a)|$ of inhabited cells takes every value in $\{1, \ldots, 7\}$. For such agents the unique type with $|\tau(a)| = 1$ is $\tau(a) = \{7\}$, realized exactly when $M_a = C_a = K_a \neq \emptyset$; the unique type with $|\tau(a)| = 7$ is $\tau(a) = \{1, \ldots, 7\}$.*

Proof. A singleton $S = \{i\}$ satisfies the three conditions of Theorem 7.2 only if the triple of R_i has all three coordinates equal to 1, i.e. $i = 7$; conversely $\{7\}$ is admissible. If $\tau(a) = \{7\}$ then $M_a = C_a = K_a = R_7^a \neq \emptyset$, and conversely. The full set is admissible. Cardinality 2 is realized by $\tau(a) = \{4, 7\}$, which meets $\{1, 4, 5, 7\}$ and $\{2, 4, 6, 7\}$ at 4 and $\{3, 5, 6, 7\}$ at 7:

take $M_a = C_a = \{x, y\}$, $K_a = \{y\}$. Cardinalities 3 to 6 are realized by $\{1, 4, 7\}$, $\{1, 2, 4, 7\}$, $\{1, 2, 3, 4, 7\}$ and $\{1, 2, 3, 4, 5, 7\}$: each contains 7 and so meets all three conditions, and the sufficiency construction of Theorem 7.2 instantiates each. □

Three sets can be drawn as three overlapping circles, and Fig. 2 does so for the seven regions; what such a drawing is and is not faithful to is fixed at Remark 4.9. Nothing below reads geometric structure off a picture: the classification above, and the results that follow from it, are statements about which cells are inhabited.

7.2. **What a Configuration Determines.**

Symbols require overlapping meaning and opacity. Symbols were placed, in Definition 4.16, in the structured ambiguity domain $\Theta = M \cap C$: an artifact is a symbol precisely when it bears meaning (M) yet retains unresolved reference (C), and only such artifacts denote (Definition 4.17). Relativized to an agent, this is an immediate unfolding of the definitions: if $\Theta_a = M_a \cap C_a = \emptyset$, then the agent-relative denotation function denotes$_a$ (Definition 5.30), whose domain is Θ_a, has empty domain, and a has no symbols. What is not immediate is which *configurations* this excludes, and the classification answers that exactly.

Corollary 7.5 (Symbol-free types)**.** *An agent a is symbol-free – $\Theta_a = \emptyset$, equivalently* denotes$_a$ *has empty domain – if and only if $4 \notin \tau(a)$ and $7 \notin \tau(a)$, i.e. $R_4^a = R_7^a = \emptyset$. Exactly* 17 *of the* 109 *types of Theorem 7.2 are symbol-free, both counts being taken under the K-objective reading declared there. Such an agent retains the cells $R_1^a, R_2^a, R_3^a, R_5^a, R_6^a$; in particular $R_2^a = C_a \setminus K_a$ and $R_6^a = C_a \cap K_a$ remain available, so opacity does not vanish with the symbols.*

Proof. By Definition 5.6, $\Theta_a = R_4^a \cup R_7^a$, which is empty iff both are. For the count, apply the criterion of Theorem 7.2 to subsets $S \subseteq \{1, 2, 3, 5, 6\}$: the conditions become $S \cap \{1, 5\} \neq \emptyset$, $S \cap \{2, 6\} \neq \emptyset$ and $S \cap \{3, 5, 6\} \neq \emptyset$, whose failures are $S \subseteq \{2, 3, 6\}$ (8 sets), $S \subseteq \{1, 3, 5\}$ (8) and $S \subseteq \{1, 2\}$ (4), with pairwise intersections of size 2 each and triple intersection of size 1; $32 - (20 - 6 + 1) = 17$. For the last clause, $M_a \cap C_a = \emptyset$ gives $R_2^a = C_a \setminus (M_a \cup K_a) = C_a \setminus K_a$ and $R_6^a = (C_a \cap K_a) \setminus M_a = C_a \cap K_a$; both involve C_a and neither is forced to be empty. □

The characterization carries a behavioural consequence, and it is the one this paper takes as principal. A query of an agent is defined in the execution layer below (Definition 8.21) as a pair whose first component denotes for that agent; the domain of denotes$_a$ is Θ_a (Definition 5.30), which the corollary above makes empty. A symbol-free agent therefore has no candidate first component, and Theorem 8.22(iii) discharges the point: such an agent poses no query at all. The seventeen types above accordingly form a *structurally query-free* subclass, and they do so under every architecture (Corollary 8.23). Two boundaries travel with the statement and are recorded at Remark 8.25: the implication runs from queries to carriers and not back, and seventeen is a lower bound on the query-free types rather than a census of them.

The extreme case among the seventeen is $\tau(a) = \{1, 2, 3\}$, where the three extensions are pairwise disjoint: $M_a \cap K_a = C_a \cap K_a = \emptyset$ as well, so the agent has neither operator cell either. It is the honest, framework-internal reading of an "agent without symbols": not an agent that has been denied symbols, but one whose extensions happen not to overlap, so

that no artifact can simultaneously mean and refer. The other sixteen symbol-free types are less extreme, differing in which operator cells they retain.

Having symbols without knowing one has them. Symbol-possession is a fact about Θ_a; whether the agent *represents* that fact is a further one. Write $\widehat{\Theta}_a \subseteq \Theta_a$ for the symbols of a that are represented as symbols in a's own self-model, and call $\Theta_a \setminus \widehat{\Theta}_a$ the agent's *symbol-unawareness*. This is a different question from the one settled in Corollary 6.6, which concerns a reading of K and not self-models, and it is not a consequence of it: the paper develops no self-model layer and takes no position, in either direction, on what an agent may establish about its own self-model. The limiting case is instructive: an agent with $\widehat{\Theta}_a = \emptyset$ but $\Theta_a \neq \emptyset$ *behaves* as though it were symbol-free, treating every artifact as a fully resolved datum, while structurally it is not – its type is one of the 92 non-symbol-free ones of Corollary 7.5. The distinction between such an agent and a genuinely symbol-free one is invisible in $\tau(a)$, which is a fact about extensions, not about self-models.

Remark 7.6 (Symbol-unawareness). We conjecture that the behavioural signature of an agent without symbols is not diagnostic of the symbol-free types of Corollary 7.5: an agent whose symbols go unrepresented in its self-model presents as one whose extensions do not overlap.

Possessing R_3, and knowing that R_3 exists. The generative-unknown cell $R_3 = K \setminus (M \cup C)$ (Definition 4.3) marks artifacts that are functional for the agent yet neither meaningful nor recognized as opaque by it. Whether an agent *has* an R_3^a cell is a fact about its type, and the classification makes the contrast sharp: of the 109 types of Theorem 7.2, 57 have $3 \in \tau(a)$ and 52 have $3 \notin \tau(a)$. These are different agents, and the difference is invisible in the extensions taken singly: it is a fact about which cells are inhabited.

Whether that fact is settled by the agent's own predicate triple is a further question, and one the paper answers for a single reading of K. Under the K-subjective reading the triple does not separate R_3^a from the absolute exterior X_{out}^a: the two differ only in the value of K_a, which that reading does not supply (Definition 5.14), so they lie undivided in the one semantic absence U_a (Lemma 5.27) and are there indistinguishable to a (Remark 5.28). Corollary 6.6 states this precisely: even when $R_3^a \neq \emptyset$ under the K-objective reading, that is not a fact about $(M_a, C_a, K_a^{\text{subj}})$. The only supplier of an objective value the framework's definitions provide is Definition 5.12, which is stated relative to a witness b with the deciding operator in R_5^b; the paper takes no position on whether other routes exist.

There is an apparent tension here that is worth naming, because the two claims are easily conflated. An agent that comes to know *that there is such a thing as* R_3 – that acquires the category "what is functional for me without being meaningful or recognized" – has thereby changed its own state: the category is an artifact of its self-model, and on the epistemogenic reading below it enters R_7^a. In that sense the cell does come to exist *for* the agent by being known. This is compatible with Corollary 6.6, which concerns something else: not whether a can hold the category, but whether a's subjective triple decides that the category is *populated*. Holding the cell as a category and settling its population are two different questions, and the corollary addresses only the second, and only under the subjective reading. This is a reading of Corollary 6.6, not a new theorem.

Self-representation creates a full triple. Finally, an agent may represent its own configuration – form an artifact standing for "my own (M, C, K) assignment". The paper conjectures that this reflexive act is *epistemogenic* (Remark 6.7): the self-artifact enters $M \cap C \cap K = R_7$ of the agent itself, because the self-model bears meaning (M), retains unresolved opacity about the agent's own hidden structure (C), and is operative on the agent's behaviour (K). On this conjecture, an agent that models its own type does not merely report $\tau(a)$; it *changes* $\tau(a)$, by bringing 7 into it. The claim is offered as a conjecture, introduces no primitive, and is used by no result above.

Remark 7.7 (A computational resource reading). Artifacts – whether functions or data – are computational. On a resource reading, an agent that instantiates the predicates M, C, K draws on three memories: (1) *storage* memory, holding the three artifact extensions – what a has integrated, what it apprehends, and what is functional for it, kept apart rather than merged (Definitions 3.2 to 3.4); (2) *execution* memory, holding what the composition of held functions consumes – the pairs of held functions that compose; and (3) *linkage* memory, holding the references among the symbols of Θ_a (Definition 5.30) together with the record of which artifacts were produced from which. The extension of each of the three is fixed later, at their single definition point in the execution layer – a forward pointer, discharged at Definition 8.27 – and the three words are used here in that extension and in no narrower one. The qualitative type $\tau(a)$ (Definition 7.1) then reads as a static snapshot of how these resources are allocated, though the snapshot withholds less than one might expect. A symbol-free agent (Corollary 7.5) keeps no record of denotation, while the other half of its linkage memory, the derivation record, is untouched by $\Theta_a = \emptyset$. An agent whose functional extension is empty holds no execution memory at all, that memory being a set of pairs drawn from K_a; under the K-subjective convention adopted later for the acting agent (Remark 8.11) emptiness of K_a is exactly the condition $5, 6, 7 \notin \tau(a)$, whereas under the K-objective reading in force in the present section it is not, since for any of the 109 types of Theorem 7.2 the absence of $5, 6, 7$ leaves $3 \in \tau(a)$ and hence $K_a = R_3^a \neq \emptyset$. The conjecture is the resource reading itself, and it does not extend to the question whether the three memories are genuinely three: that question is settled below, and settled affirmatively, linkage being provably not determined by storage (Definition 8.27 and Corollary 8.29). The reading is computational and resource-level; it introduces no temporal claim, and how these memories evolve is the province of an execution dynamics treated elsewhere.

Remark 7.8 (Collected: what an absent cell withholds). The results of this section and of Section 6.2 are collected here in one direction of reading. Nothing below is new: each item restates a definition, a result, or – in item (iii) – a conjecture already in place, carrying here exactly the status it carries there, and nothing below is proved here. The tag above is accordingly the status of items (i) and (ii); item (iii) inherits the conjectural status of Remark 7.7 and is marked as such where it is stated.

(i) A cell absent from $\mathrm{Cap}(a)$ (Definition 6.4) is a limitation of a's classificatory and operational reach, and the absence is a property of a rather than of the artifact domain (Remark 6.5).

(ii) When the absent cells are the two symbol cells – $4, 7 \notin \tau(a)$ – what is withheld is symbol-bearing itself: $\Theta_a = \emptyset$ and $\mathrm{denotes}_a$ has empty domain (Corollary 7.5), and with it the capacity to pose a question, under every architecture (Corollary 8.23); the direction of that implication and the scope of the count are as recorded above. *Whether* an agent bears symbols is thus fixed by which cells it inhabits, Corollary 7.5 being an

equivalence. Which kinds of symbol it may bear is a different question, and not one this paper raises: no taxonomy of symbol kinds is developed here (Remark 4.18), and nothing in this remark bears on one.

(iii) On the resource reading of Remark 7.7 – which is offered there as a conjecture – an absent cell can also be read as something the agent need not hold, taken in the extension fixed at Definition 8.27 and with the qualifications Remark 7.7 attaches to it. A symbol-free agent keeps no record of denotation, while the derivation half of its linkage memory is untouched. An agent with $5, 6, 7 \notin \tau(a)$ holds no execution memory under the K-subjective convention adopted later for the acting agent (Remark 8.11), where those absences leave $K_a = \emptyset$; under the K-objective reading declared in Theorem 7.2 the same absences leave $3 \in \tau(a)$ for every one of the 109 types there, so K_a is non-empty and no such conclusion follows. Linkage is in either case provably not recoverable from storage (Corollary 8.29).

7.3. **Type-Determined Structural Profiles.** The type $\tau(a)$ records which cells an agent inhabits. Several questions asked of an agent turn only on coarser information – whether it bears a symbol at all, whether it operates anything, whether it holds inert content. This subsection isolates that coarser datum, characterizes exactly which of its values occur, and records the order it carries.

Definition 7.9 (Structural profile)**.** For an admissible type τ (Theorem 7.2) set

$$\begin{aligned} \Gamma(\tau) \;&=\; (\mathrm{SYM}, \mathrm{OPS}, \mathrm{TRN}, \mathrm{OPQ}, \mathrm{EPI}, \mathrm{INR}) \;\in\; \{0,1\}^6, \\ \mathrm{SYM}(\tau) &= [\,\tau \cap \{4,7\} \neq \emptyset\,], \quad \mathrm{OPQ}(\tau) = [\,6 \in \tau\,], \\ \mathrm{OPS}(\tau) &= [\,7 \in \tau\,], \qquad\qquad \mathrm{EPI}(\tau) = [\,3 \in \tau\,], \\ \mathrm{TRN}(\tau) &= [\,5 \in \tau\,], \qquad\qquad \mathrm{INR}(\tau) = [\,\tau \cap \{1,2\} \neq \emptyset\,]. \end{aligned}$$

The coordinates record, in order, that the agent bears a symbol ($\Theta_a \neq \emptyset$, Definition 4.16 and Corollary 7.5); that it bears one which is also functional (R_7); that it holds a transparent operator (R_5); an opaque one (R_6); that its pure-functionality cell is inhabited (R_3); and that it holds non-functional content (R_1 or R_2). Γ is a function of τ and of nothing else, so agents of equal type have equal profile. We reserve *capability profile* for τ itself (Definition 7.1) and call $\Gamma(\tau)$ the *structural profile*.

Lemma 7.10 (Symbol-operability entails symbol-bearing)**.** $\mathrm{OPS}(\tau) = 1$ *implies* $\mathrm{SYM}(\tau) = 1$.

Proof. $7 \in \tau$ gives $\tau \cap \{4,7\} \neq \emptyset$. Structurally this is the inclusion $R_7 \subseteq \Theta$ of Definition 4.16. □

Lemma 7.11 (Admissibility in profile coordinates)**.** *Every admissible τ satisfies*

$$\begin{aligned} &\text{(K)} \quad \mathrm{EPI} \vee \mathrm{TRN} \vee \mathrm{OPQ} \vee \mathrm{OPS}, \\ &\text{(M)} \quad \mathrm{INR} \vee \mathrm{SYM} \vee \mathrm{TRN}, \\ &\text{(C)} \quad \mathrm{INR} \vee \mathrm{SYM} \vee \mathrm{OPQ}. \end{aligned}$$

Proof. By Theorem 7.2 an admissible τ meets $\{3,5,6,7\}$, $\{1,4,5,7\}$ and $\{2,4,6,7\}$. The cells $3, 5, 6, 7$ are recorded by $\mathrm{EPI}, \mathrm{TRN}, \mathrm{OPQ}, \mathrm{OPS}$ respectively, giving (K). For (M): cell 1 contributes INR, cells 4 and 7 contribute SYM, cell 5 contributes TRN. For (C): cell 2 contributes INR, cells 4 and 7 contribute SYM, cell 6 contributes OPQ. □

Theorem 7.12 (Image of Γ)**.** *A profile $\gamma \in \{0,1\}^6$ lies in* $\operatorname{Im}\Gamma$ *if and only if it satisfies* $\mathrm{OPS} \Rightarrow \mathrm{SYM}$ *together with* (K)*,* (M) *and* (C) *of Lemma 7.11.*

Proof. Necessity is Lemmas 7.10 and 7.11. For sufficiency, let γ satisfy the four conditions and put

$$\tau \;=\; \{7 \mid \mathrm{OPS}\} \cup \{4 \mid \mathrm{SYM} \wedge \neg\mathrm{OPS}\} \cup \{5 \mid \mathrm{TRN}\} \cup \{6 \mid \mathrm{OPQ}\} \cup \{3 \mid \mathrm{EPI}\} \cup \{1,2 \mid \mathrm{INR}\}.$$

Then $\Gamma(\tau) = \gamma$: the coordinates $\mathrm{OPS}, \mathrm{TRN}, \mathrm{OPQ}, \mathrm{EPI}$ read single cells placed exactly when required; SYM holds because 7 enters when OPS and 4 enters when $\mathrm{SYM} \wedge \neg\mathrm{OPS}$; and INR holds because 1 and 2 enter together exactly when INR. For admissibility, (K) places one of $3, 5, 6, 7$ in τ; (M) places 1, or 4 or 7, or 5; and (C) places 2, or 4 or 7, or 6. Taking *both* 1 and 2 for INR is what lets a single disjunct discharge (M) and (C) independently. □

Corollary 7.13 (Profile census)**.** $|\operatorname{Im}\Gamma| = 39$ *of the* 64 *profiles; the* 25 *excluded are exactly those failing one of the four conditions of Theorem 7.12, ten of them by* $\mathrm{OPS} \Rightarrow \mathrm{SYM}$ *alone.*

Proof. A finite check of the four conditions over $\{0,1\}^6$, which Theorem 7.12 converts into a statement about the image. □

Proposition 7.14 (Order structure)**.** *Order profiles coordinatewise. Then* $\operatorname{Im}\Gamma$ *is a finite join-semilattice with greatest element* $(1,1,1,1,1,1)$*. It is not closed under meet and has no least element.*

Proof. The conditions (K), (M) and (C) are disjunctions of coordinates and so survive coordinatewise join. For $\mathrm{OPS} \Rightarrow \mathrm{SYM}$: if the join has $\mathrm{OPS} = 1$ then some operand does, that operand has $\mathrm{SYM} = 1$, and hence so does the join. By Theorem 7.12 the join therefore lies in the image, and the all-ones profile satisfies the four conditions and dominates every member. For the failure of meet, write a profile by its active coordinates: $(\mathrm{TRN}, \mathrm{INR})$ and $(\mathrm{TRN}, \mathrm{OPQ})$ both satisfy the conditions, while their meet (TRN) has $\mathrm{INR} = \mathrm{SYM} = \mathrm{OPQ} = 0$ and fails (C). A least element would lie below both, hence below that meet, leaving only (TRN) and the all-zero profile, and neither is in the image. □

Remark 7.15 (The join structure is forced by the shape of the constraints). The preceding proof is not an accident of these particular six coordinates, and it is worth saying why, since the reason also shows that the proposition carries less independent weight than its statement suggests. Read the four conditions of Theorem 7.12 as a conjunctive normal form. Three of them are disjunctions of coordinates with no negation, and the fourth, $\mathrm{OPS} \Rightarrow \mathrm{SYM}$, is the disjunction $\neg\mathrm{OPS} \vee \mathrm{SYM}$: every clause carries at most one negated coordinate. A Boolean relation definable by such a normal form is closed under coordinatewise join, and the closure argument above is exactly the general one specialised to these clauses. The greatest element is then automatic rather than a separate observation: a non-empty finite join-closed subset of $\{0,1\}^6$ contains the join of all its members. So Proposition 7.14 is not independent of Theorem 7.12 – the semilattice structure is a consequence of the *form* of the exact characterization, and the failure of meet is what has to be checked separately, as it was.

Remark 7.16 (What the structural profile does not classify). Γ is a function of τ, so it classifies exactly what τ fixes, and three kinds of datum lie outside it. The extensions themselves are one: Θ_a as a set, and the triple $\mathrm{Sto}(a)$ (Definition 8.27), are fixed by the extensions rather than by the type, since agents of equal type may inhabit the same cells with different artifacts. The architecture is a second: Exe is fixed by the extensions together

with $\circ_a$, and Theorem 8.30 exhibits agents of equal type whose Exe differs. Data beyond either make a third. Lnk is of that kind: the same theorem separates it at equal type, but its denotation component is $\mathrm{denotes}_a \restriction \Theta_a$, and Definition 5.30 leaves $\mathrm{denotes}_a$ a datum of the agent, so the architecture does not fix it either. Whether a query is available is likewise of that kind, turning on $\mathrm{denotes}_a$ and on a region assignment (Remark 8.25). One consequence does run in the direction the type fixes: $\mathrm{SYM}(\tau) = 0$ holds for exactly the 17 types of Corollary 7.5, and those admit no query under any architecture (Corollary 8.23). Γ accordingly classifies type-determined structural presence, and not the extensional or operational state of an agent.

8. Execution: Generation and Region Movement

The classification developed so far is complete but inert: it says where an artifact stands for an agent, not what an agent does. This section supplies the missing half, and it does so without leaving the static setting. Its subject is a *relation* between region assignments, and nothing temporal: no time index and no cost model is introduced, and the word *step* below is a relation symbol, on exactly the reading under which the closure of Definition 8.2 stratifies K_α by derivation order rather than by execution. The finite sequences of generation steps quantified over in Theorem 8.32 are derivation orders in that same sense, not temporal runs, and no step is dated or costed. What the section fixes is which pairs (source cell, transmutation) are coherent (Theorem 8.17 and Proposition 8.19), in which regions internally produced artifacts land (Theorem 8.16), what a question is as a formal object (Definition 8.21 and Theorem 8.22), which of an agent's records are recoverable from its artifact extensions (Corollary 8.29), and where internal generation reaches its fixed point (Theorem 8.32). Throughout, the acting agent's own assignment is read K-subjectively (Remark 8.11).

8.1. Agent Architecture and Closure.

Here $\mathcal{A}$ is the artifact domain (Definition 3.1) and $\mathcal{A}^+$ the classified domain (Definition 3.7).

Definition 8.1 (Compositional Architecture)**.** A *compositional architecture* over $\mathcal{A}$ is a pair

$$\alpha = (\mathcal{O},\ \circ)$$

where:

(1) $\mathcal{O} \subseteq \mathcal{A}$ is the *operational basis*;
(2) $\circ\colon \mathcal{A} \times \mathcal{A} \rightharpoonup \mathcal{A}$ is a partial binary *composition operation*;
(3) α is a meta-level structure: $\alpha \notin \mathcal{A}$.

No algebraic axioms (associativity, identity, commutativity) are assumed unless explicitly stated. The basis is *not* required to be non-empty: the empty basis is admissible and generates the empty closure. Admitting it costs nothing here: the results that need a non-empty basis carry a non-triviality hypothesis (Definition 8.3), which forces $\mathcal{O} \neq \emptyset$ exactly where it is required.

The composition $\circ$ is declared on $\mathcal{A} \times \mathcal{A}$ but its effective domain is restricted by the inductive closure below.

Definition 8.2 (Compositional Closure)**.** Given a compositional architecture $\alpha = (\mathcal{O}, \circ)$ over $\mathcal{A}$, the *generation sequence* $(K_\alpha^{(n)})_{n\geq 0}$ is defined inductively:

$$K_\alpha^{(0)} = \mathcal{O}, \tag{8.1}$$

$$K_\alpha^{(n+1)} = K_\alpha^{(n)} \cup \{\, f \circ g \mid f, g \in K_\alpha^{(n)},\ f \circ g \text{ is defined} \,\}. \tag{8.2}$$

The *compositional closure* induced by α is

$$K_\alpha = \bigcup_{n\geq 0} K_\alpha^{(n)}.$$

The sequence is monotonically non-decreasing: $K_\alpha^{(n)} \subseteq K_\alpha^{(n+1)}$ for all n. For finite $\mathcal{A}$, it stabilises in finitely many steps.

Definition 8.3 (Genesis architecture: depth, non-triviality, stabilization, depth class)**.** Let $\alpha = (\mathcal{O}, \circ)$ be a compositional architecture with induced closure K_α (Definition 8.2).

(1) **Generation depth.** For $x \in K_\alpha$, $d_\alpha(x) = \min\{\, n \in \mathbb{N} \mid x \in K_\alpha^{(n)} \,\}$. Elements of $\mathcal{O}$ have depth 0; an element x has depth $n \geq 1$ iff $x \in K_\alpha^{(n)} \setminus K_\alpha^{(n-1)}$.
(2) **Non-triviality.** α is *non-trivial* if $K_\alpha \supsetneq \mathcal{O}$, i.e. there exists $x \in K_\alpha$ with $d_\alpha(x) \geq 1$.
(3) **Stabilization depth.** $n^*(\alpha) = \min\{\, n \in \mathbb{N} \mid K_\alpha^{(n+1)} = K_\alpha^{(n)} \,\}$. The minimum exists exactly when the generation sequence stabilises, guaranteed whenever $\mathcal{A}$ is finite; then $n^*(\alpha) \leq |K_\alpha| - |\mathcal{O}|$.
(4) **Depth class.** The *depth-n functional class* is $\mathcal{F}_\alpha^{(n)} = \{\, x \in K_\alpha \mid d_\alpha(x) = n \,\}$; the family $\{\mathcal{F}_\alpha^{(n)}\}_{n=0}^{n^*(\alpha)}$ partitions K_α.

Definition 8.3 introduces the internal stratification of K_α that is absent from the extensional predicate K of Definition 3.4. The following two definitions extend the classification to the full classified domain $\mathcal{A}^+$ and to the structural phenomena enabled by composition.

Definition 8.4 (Bigraded Classification)**.** Under a compositional architecture α, every artifact $x \in \mathcal{A}^+$ is classified by the pair (R_i, d) where $R_i \in \{R_1, \ldots, R_7\}$ is the MCK region (Section 4) and

$$d = \begin{cases} d_\alpha(x) & \text{if } x \in K_\alpha, \\ \bot & \text{if } x \notin K_\alpha. \end{cases}$$

Definition 8.5 (Minimal Generating Set)**.** A subset $\mathcal{O}_{\min} \subseteq \mathcal{O}$ is a *minimal generating set* for K_α under $\circ$ if:

(1) $\mathrm{cl}_\circ(\mathcal{O}_{\min}) = K_\alpha$;
(2) for every $o \in \mathcal{O}_{\min}$: $\mathrm{cl}_\circ(\mathcal{O}_{\min} \setminus \{o\}) \subsetneq K_\alpha$.

Here $\mathrm{cl}_\circ(S)$ denotes the compositional closure of S under $\circ$, constructed as in Definition 8.2 with $\mathcal{O}$ replaced by S.

Remark 8.6 (Optional Algebraic Axioms). The following axioms may be imposed on specific architectures when needed:

COMP-A1 (Associativity): For all $f, g, h \in K_\alpha$ where all compositions are defined: $(f \circ g) \circ h = f \circ (g \circ h)$.
COMP-A2 (Identity): There exists $e \in \mathcal{O}$ such that $e \circ f = f \circ e = f$ for all $f \in K_\alpha$ where the compositions are defined.

Neither axiom is required by the results of this section.

Lemma 8.7 (Generic indistinguishability)**.** *Let $M, C \subseteq \mathcal{A}$ be arbitrary and let $K \subseteq \mathcal{A}$ with $|K| \geq 2$. There exist architectures $\alpha_1 \neq \alpha_2$, each with basis contained in K, such that $K_{\alpha_1} = K_{\alpha_2} = K$ and $d_{\alpha_1} \neq d_{\alpha_2}$. The two systems so obtained carry the* same *triple (M, C, K).*

Proof. Let $K = \{a_1, \ldots, a_n\}$ with $n \geq 2$.

Architecture α_1: $\mathcal{O}_1 = K$, $\circ_1$ nowhere defined. Then $K_{\alpha_1} = K$ and $d_{\alpha_1}(x) = 0$ for all $x \in K$.

Architecture α_2: $\mathcal{O}_2 = K \setminus \{a_n\}$, $\circ_2$ defined by $a_1 \circ_2 a_1 = a_n$; all else undefined. Then $K_{\alpha_2}^{(0)} = K \setminus \{a_n\}$ and $K_{\alpha_2}^{(1)} = K_{\alpha_2}^{(0)} \cup \{a_n\} = K$. Hence $K_{\alpha_2} = K$, but $d_{\alpha_2}(a_n) = 1 \neq 0 = d_{\alpha_1}(a_n)$.

Both bases are contained in K, so both architectures are admissible in the sense of Definition 8.12. The construction varies only $(\mathcal{O}, \circ)$ and leaves M and C untouched, so the two systems carry the same triple (M, C, K); the static classification of Section 4 therefore assigns every artifact the same region in both, and the two are indistinguishable to it. The compositional theory distinguishes them via the depth function. □

The indistinguishability is generic: it holds for *every* non-singleton K, not for a specific small witness. Treating K as primitive therefore discards, for every non-trivial functional set, distinctions that the compositional language can express.

The lemma has a consequence about the closure operator itself which is worth stating separately, since the closure is the object the generative layer converges to (Theorem 8.32) and one might expect it to carry the record of how it was reached.

Corollary 8.8 (The closure does not determine the stratification)**.** *There is no function F with $d_\alpha = F(K_\alpha)$ for every compositional architecture α, and none with $\{(f, g) \mid f \circ_\alpha g \text{ defined}\} = F(K_\alpha)$ either. Equivalently, two architectures may induce the same compositional closure (Definition 8.2) while differing in the generation depth of its elements (Definition 8.3) and in the pairs on which their compositions are defined.*

Proof. Take any $K \subseteq \mathcal{A}$ with $|K| \geq 2$ and let α_1, α_2 be the architectures of Lemma 8.7. They satisfy $K_{\alpha_1} = K_{\alpha_2} = K$, so a function of the closure alone would have to assign them the same value; but $d_{\alpha_1} \neq d_{\alpha_2}$ there, which refutes the first claim. For the second, $\circ_1$ is nowhere defined while $a_1 \circ_2 a_1 = a_n$, so the two sets of defined pairs are $\emptyset$ and a non-empty set. □

Remark 8.9 (The hypothesis is sharp). The bound $|K| \geq 2$ is not a technical convenience but the exact threshold. If $K_\alpha = \emptyset$ then $\mathcal{O} = \emptyset$, since the generation sequence starts at $\mathcal{O}$ and can only grow; and if $K_\alpha = \{x\}$ then $\mathcal{O} = \{x\}$ for the same reason, so $d_\alpha(x) = 0$ whatever $\circ$ does. In both cases the closure determines the depth function outright, and no separation is available. Generative ambiguity therefore appears exactly at the first size that admits it.

Remark 8.10 (What the corollary does and does not say). The datum shown not to be recoverable is the depth stratification of Definition 8.3, a function $K_\alpha \to \mathbb{N}$, together with the domain of the composition. It is not a derivation history: nothing here records which pair produced a given artifact, and the corollary should not be read as a statement about provenance in that fuller sense. The record of which pair produced which artifact is a separate datum, carried by Lnk (Definition 8.27), and its own irrecoverability is the content of Corollary 8.29.

8.2. **The Step Relation.**

Remark 8.11 (K-reading convention for the acting agent). Throughout this section the acting agent's own assignment is read under K-subjective (Definition 5.14), so that $R_3^a = \emptyset$ for the acting agent and $K_a(x) = 1$ implies $M_a(x) = 1$ or $C_a(x) = 1$; the K-objective reading is what an *external* witness b uses when it classifies a's artifacts (Definition 5.12 and Proposition 5.15). This implication is a property of the subjective *reading* of K, fixed in Definition 5.14, and not a violation of Axiom 3.12, whose scope is the primitive predicates over the artifact domain and which continues to deny each of the four implications it displays – none of which is the disjunctive one at issue here.

An agent just is a predicate triple (Definition 5.1), and a triple just is a region assignment (Definition 5.6). The relation below therefore relates two agents carrying the same index, the same operational basis and the same composition; nothing is asserted about order, and no state beyond the assignment is introduced. We identify each cell with its predicate triple in $\{0,1\}^3$ and each region assignment $\mathrm{Region}_a(x)$ with that triple, as licensed by Proposition 4.20.

Definition 8.12 (Agent architecture and the step relation)**.** Let a be an agent (Definition 5.1). An *agent architecture* for a is a compositional architecture $\alpha_a = (\mathcal{O}_a, \circ_a)$ over $\mathcal{A}$ (Definition 8.1) with $\mathcal{O}_a \subseteq K_a$: this is the agent-relative instantiation of Definition 8.1, exactly as Definition 5.6 is the agent-relative instantiation of Definition 3.7, and it introduces no new primitive. An *execution step* of a relates two region assignments ρ, ρ' of a that agree off a single artifact and is of one of three kinds:

GEN: there are $f, h \in K_a$ – the extensions being read under ρ – with $f \circ_a h$ defined and $y = f \circ_a h$ such that $K_a(y) = 0$ under ρ, the output not yet being functional for a, and $K_a(y) = 1$ under ρ': the step is a's acquisition of a function it did not hold;

MOV: there is x with $\rho(x) \neq (0,0,0)$ and $\rho'(x) = v \neq \rho(x)$, $v \neq (0,0,0)$, together with a *license*: an artifact $z \in \mathcal{A}_a^+$ produced by a GEN step of a whose content is a certificate of understanding for x (Definition 3.2), a refutation of one, or an execution witness for x;

EXIT: there is x with $\rho(x) \neq (0,0,0)$ and $\rho'(x) = (0,0,0)$: the artifact leaves a's classified domain, the target being the absolute exterior cell (Remark 3.9), under a license as above.

For a MOV or EXIT step affecting x we write $x\colon u \rightsquigarrow_a v$ with $u = \rho(x)$, $v = \rho'(x)$, and $\rightsquigarrow_a^*$ for the reflexive-transitive closure of $\rightsquigarrow_a$. A step is *realizable by* a if a holds the artifacts its kind requires: the operands f, h for GEN, the license z for MOV and EXIT. What makes a given z a license is not formalized here – the framework carries no certificate calculus – and the licensing clause is a stipulation of the step relation, not a derived condition.

Two scope facts about this relation are best stated at once, because a good deal below depends on them. First, a step holds the architecture $\alpha_a = (\mathcal{O}_a, \circ_a)$ fixed: the basis and the composition are the same on both sides of every step, and no kind of step alters either. Second, the artifacts a step requires the agent to *hold* come from within: GEN reads its operands off K_a and returns their composite, and the license required by MOV and EXIT is itself the output of a GEN step of a. The relation therefore contains no operation of *supply*: it provides no step by which an externally furnished artifact enters a's classified domain, and none by which such an artifact could serve as a license. Acquisition is not modelled here at all. This is a boundary of the layer, not a thesis about agents, and the results below are to

be read inside it: where one of them stops at the closure of a basis, what it establishes is a fact about a fixed architecture and not a necessity about anything outside one.

Remark 8.13 (Generation is not a group element). GEN is typed as a distinct kind of step, and deliberately not as the application of a transmutation (Proposition 4.20) to a source cell. Its source may be the absolute exterior, which Remark 3.9 forbids treating as a source cell at all; and even from an active cell the step is not determined by a group element, its precondition constraining the *operands* f, h rather than the cell of y and its effect on that cell being fixed only up to Axiom 8.14. Only MOV and EXIT carry a transmutation, which is why $\rightsquigarrow_a$ is written for those two alone.

Axiom 8.14 (Output apprehension)**.** If agent a realizes a GEN step producing y (Definition 8.12), $M_a(y) = 1$ or $C_a(y) = 1$ holds in the resulting assignment.

The stipulation reads off the registration clause of Definition 3.3: $C(y) = 1$ records that y is recognized as an artifact – registered in the ontology – with its resolution incomplete, and producing an artifact registers it. An agent's own output is therefore at least apprehended; whether it is also resolved ($M_a(y) = 1$) or leaves a remainder ($C_a(y) = 1$) is not decided by the axiom, and both may hold (Axiom 3.11). This does not contradict the three predicate axioms of Section 3.4: those deny implications among the predicates over *arbitrary* artifacts, whereas Axiom 8.14 is conditional on a hypothesis outside the predicate language – that y is the output of a realized GEN step of a.

Remark 8.15 (Unregistered by-products). We conjecture that an agent may cause an artifact to exist without registering it, and that such by-products fall outside Axiom 8.14, whose hypothesis is a *realized* GEN step, i.e. one whose output the agent holds. For a itself such a by-product falls in the semantic absence U_a, since $M_a = C_a = 0$ (Definition 5.26 and Lemma 5.27), and under the subjective reading of K it is there indistinguishable from the absolute exterior (Remark 5.28). The conjecture introduces no primitive and is used by no result below.

Theorem 8.16 (Generation lands in the active regions)**.** *Every artifact y produced by a realized GEN step of a occupies, for a, one of the six active regions $R_1^a, R_2^a, R_4^a, R_5^a, R_6^a, R_7^a$. In particular no GEN step of a populates R_3^a, and none leaves y in the absolute exterior.*

Proof. By Axiom 8.14, $M_a(y) = 1$ or $C_a(y) = 1$, so $y \in \mathrm{KNOWN}_a$ (Definitions 3.5 and 5.8) and hence $y \in \mathcal{A}_a^+$. By Proposition 5.7, y occupies exactly one relativized region. The cells excluded are the two with $M_a(y) = C_a(y) = 0$: the absolute exterior $(0,0,0)$ and $R_3^a = (0,0,1)$ (Proposition 3.8 applied to (M_a, C_a, K_a)). The six remaining cells are the active regions. Sharper: Definition 8.12 also puts $K_a(y) = 1$, so y occupies R_5^a, R_6^a or R_7^a; the six-region form is what the results below consume. □

The payoff is a change of status for the suspension. Axiom 4.11 calls the emptiness of R_3 a scoping convention rather than a theorem, and Remark 4.12 is explicit that it is not derived. For *internally generated* artifacts it now is derived – relative to Axiom 8.14, which the execution layer stipulates – and derived under the reading that makes the statement non-vacuous: under K-subjective the exclusion of R_3^a is already carried by Remark 8.11, but under the K-objective reading an external witness uses, R_3^a may be inhabited (Proposition 5.15) and Theorem 8.16 is a genuine restriction on where a's own outputs can lie. Under either reading the conclusion is the same: R_3^a is inhabited only by artifacts a did not produce.

8.3. **Which Transmutations an Agent Can Realize.** A MOV or EXIT step carries a transmutation: by simple transitivity (Proposition 4.20(i)) the step $x\colon u \rightsquigarrow_a v$ determines the unique $g = u \oplus v \in \mathbb{Z}_2^3$ toggling exactly the predicates on which u and v differ. The question this subsection settles is which transmutations an agent can apply, and from which cells.

Theorem 8.17 (Stabiliser of the active cells)**.** *Let*

$$\mathcal{S} \;=\; \{0,1\}^3 \setminus \{(0,0,0),\,(0,0,1)\}$$

be the six active cells of Section 4.3, read for the acting agent per Remark 8.11, and let

$$H(\mathcal{S}) \;=\; \{\, g \in \mathbb{Z}_2^3 \mid g \oplus \mathcal{S} = \mathcal{S} \,\}$$

be its setwise stabiliser under the transmutation action of Proposition 4.20. The following hold:

(i) $H(\mathcal{S}) = \{\, g \mid u \oplus g \in \mathcal{S}$ *for every* $u \in \mathcal{S} \,\}$*: the stabiliser is exactly the set of transmutations coherent from* every *cell the agent can occupy;*

(ii) $H(\mathcal{S}) = \{(0,0,0),\,(0,0,1)\} = \langle (0,0,1) \rangle \cong \mathbb{Z}_2$*, the subgroup generated by the pure functionality toggle;*

(iii) *under the K-objective reading, where only* $(0,0,0)$ *is excluded and the active set is* $\mathcal{S}^{\mathrm{obj}} = \{0,1\}^3 \setminus \{(0,0,0)\}$*, the corresponding stabiliser* $H(\mathcal{S}^{\mathrm{obj}})$ *is trivial.*

Proof. (i) Translation by g is a bijection of $\{0,1\}^3$ (Proposition 4.20(i)), so $g \oplus \mathcal{S}$ has the same finite cardinality as $\mathcal{S}$; an inclusion $g \oplus \mathcal{S} \subseteq \mathcal{S}$ between finite sets of equal size is an equality.

(ii) Since translation by g is a bijection of the whole cube, it maps $\mathcal{S}$ onto $\mathcal{S}$ if and only if it maps the complement $E = \{0,1\}^3 \setminus \mathcal{S}$ onto E. Here $E = \{(0,0,0),(0,0,1)\}$, which is a *subgroup* of $\mathbb{Z}_2^3$: it contains the identity and is closed, as $(0,0,1) \oplus (0,0,1) = (0,0,0)$. For a subgroup E and any $g \in \mathbb{Z}_2^3$ one has $g \oplus E = E$ if and only if $g \in E$: closure gives the implication from left to right of the membership, and conversely $g = g \oplus (0,0,0) \in g \oplus E = E$. Hence $H(\mathcal{S}) = E = \langle (0,0,1) \rangle$, of order 2.

(iii) The same argument applies with $E = \{(0,0,0)\}$, the trivial subgroup, giving $H(\mathcal{S}^{\mathrm{obj}}) = \{(0,0,0)\}$. □

Read operationally, this is the section's central statement: *the only non-identity transmutation available from every region the agent can occupy is the one along the functionality axis; every semantic transmutation is conditioned on where the artifact already stands.* Execution is free along K and always conditioned along M and C. The stabiliser $H(\mathcal{S})$ is a subgroup of transmutations and is unrelated to the stabilization depth $n^*(\alpha)$ of Definition 8.3, which is an integer; no notation is shared.

Remark 8.18 (The stabiliser is a symmetry, not a coincidence). Theorem 8.17(ii) is an instance of a general fact and not a numerical accident of the pure-K cell. For any cell $c \neq (0,0,0)$, the two-element set $\{(0,0,0), c\}$ is a subgroup of $\mathbb{Z}_2^3$, since $c \oplus c = (0,0,0)$; by the argument of Theorem 8.17(ii), suspending c alongside the absolute exterior leaves a stabiliser equal to $\langle c \rangle$, of order 2. Suspending *any* single region therefore preserves exactly the toggle that generates that region, and nothing else. The framework suspends $R_3 = (0,0,1)$ (Axiom 4.11), so the toggle it preserves is the K-toggle. Had the pure-understanding cell $R_1 = (1,0,0)$ been suspended instead, the preserved toggle would be the M-toggle. The content of Theorem 8.17 is thus the *choice* of suspended cell, not the existence of a preserved toggle.

This fixes the cost of admitting external witnesses. Admitting the K-objective reading for the acting agent – equivalently, admitting external witnesses in the sense of Definition 5.12 – reactivates R_3^a (Proposition 5.15) and destroys the universality of the K-toggle: by Theorem 8.17(iii) the stabiliser collapses to the identity, so no non-identity transmutation is coherent from every occupiable cell. Concretely, the K-toggle applied to the reactivated source $R_3^a = (0,0,1)$ has target $(0,0,0)$, the absolute exterior, so it is an EXIT rather than a MOV from that source. The trade-off is exact, and it is worth stating in one sentence: external witnessing buys reachability – the seventh cell becomes occupiable, and Corollary 5.25 records the resulting gain from six to seven distinct assignments – and pays for it with uniformity, since no transmutation remains coherent everywhere. This is the hinge into Remark 8.20 below: the pairs no step of a can carry out are exactly those whose target is the cell only an external witness's classification can certify.

8.3.1. *Admissibility is a Property of Pairs.*

Proposition 8.19 (Pairwise admissibility)**.** *Work under Remark 8.11, and call a pair* $(u,g) \in \mathcal{S} \times \mathbb{Z}_2^3$ admissible *if* $u \oplus g \in \mathcal{S}$, blocked *if* $u \oplus g = (0,0,1)$, *and* exiting *if* $u \oplus g = (0,0,0)$. *The following hold:*

(i) *each of the six elements* $g \notin H(\mathcal{S})$ *is admissible from exactly four active sources, blocked from exactly one, namely* $u = g \oplus (0,0,1)$, *and exiting from exactly one, namely* $u = g$;
(ii) *each of the two elements of* $H(\mathcal{S})$ *is admissible from all six active sources, with no blocked and no exiting source;*
(iii) *of the* 48 *pairs in* $\mathcal{S} \times \mathbb{Z}_2^3$, *exactly* 36 *are admissible,* 6 *are blocked and* 6 *are exiting;*
(iv) *dually, every active source* u *has exactly one blocked element,* $g = u \oplus (0,0,1)$, *and exactly one exiting element,* $g = u$.

Proof. Fix g. By Proposition 4.20(i) the equation $u \oplus g = w$ has the unique solution $u = g \oplus w$, so g has at most one exiting source, $u = g \oplus (0,0,0) = g$, and at most one blocked source, $u = g \oplus (0,0,1)$. The exiting source is active exactly when $g \in \mathcal{S}$, i.e. when $g \notin H(\mathcal{S}) = \{(0,0,0),(0,0,1)\}$ (Theorem 8.17(ii)); the blocked source is active exactly when $g \oplus (0,0,1) \in \mathcal{S}$, i.e. when $g \oplus (0,0,1) \notin \{(0,0,0),(0,0,1)\}$, which again says $g \notin H(\mathcal{S})$. For $g \notin H(\mathcal{S})$ the two sources are distinct, as they differ by $(0,0,1) \neq (0,0,0)$, and the remaining $6-2=4$ active sources are admissible, proving (i). For $g \in H(\mathcal{S})$ neither source is active, so all six are admissible, proving (ii) and recovering Theorem 8.17(i). Counting, $6 \cdot 4 + 2 \cdot 6 = 36$ admissible, $6 \cdot 1 = 6$ blocked and $6 \cdot 1 = 6$ exiting pairs, totalling $48 = 6 \cdot 8$, which is (iii). Part (iv) is the same uniqueness applied with u fixed instead of g: $u \oplus g = (0,0,1)$ has the unique solution $g = u \oplus (0,0,1)$ and $u \oplus g = (0,0,0)$ the unique solution $g = u$. □

Table 1: Admissibility of the eight transmutations of Proposition 4.20 over the six active source cells, under the reading of Remark 8.11. The blocked source is the one whose target is the suspended cell R_3^a; the exit source is the one whose target is the absolute exterior. Totals: 36 admissible, 6 blocked, 6 exiting pairs (Proposition 8.19).

g	active sources admitting g	blocked	exit	reading
$(0,0,0)$	$R_1, R_2, R_4, R_5, R_6, R_7$	none	none	identity: no change
$(0,0,1)$	$R_1, R_2, R_4, R_5, R_6, R_7$	none	none	functionality toggle
$(1,0,0)$	R_2, R_4, R_6, R_7	R_5	R_1	meaning toggle
$(0,1,0)$	R_1, R_4, R_5, R_7	R_6	R_2	opacity toggle
$(1,1,0)$	R_1, R_2, R_5, R_6	R_7	R_4	semantic swap
$(1,0,1)$	R_2, R_4, R_6, R_7	R_1	R_5	meaning and functionality
$(0,1,1)$	R_1, R_4, R_5, R_7	R_2	R_6	opacity and functionality
$(1,1,1)$	R_1, R_2, R_5, R_6	R_4	R_7	full toggle

Table 1 lists the eight cases. The result to extract is not the arithmetic but its typing consequence, and we state it plainly: the question *which of the eight transmutations are admissible* is ill-typed at the level of group elements. Six of the eight are admissible from four sources, blocked from one and exiting from one; only the two elements of the stabiliser admit a source-free answer. Admissibility is a property of the *pair* (source cell, transmutation), not of the transmutation. This is a result about the shape of the question, not an evasion of it: Proposition 8.19 answers the well-typed version completely, for all 48 pairs. Admissibility is a condition on the pair alone, and hence a fact about the group action; whether an admissible pair is carried out by a step of a turns in addition on the licensing clause of Definition 8.12, which is no part of that arithmetic.

Remark 8.20 (The blocked class, and what would settle it). The six blocked pairs of Proposition 8.19 are exactly the pairs whose target is R_3^a, one for each active source (Proposition 8.19(iv)). No step of a carries any of them out: under Remark 8.11 the acting agent has $R_3^a = \emptyset$, so no MOV of a's own assignment has a target there, EXIT targets the absolute exterior by Definition 8.12, and no GEN step populates R_3^a either (Theorem 8.16). Supply is not a further case to be argued about: as recorded after Definition 8.12, the step relation has no supply operation, so what an agent might receive from outside is not in the scope of this proposition at all. What the blocked class turns on is instead the reading of K. Membership in R_3^a is a fact under the K-objective reading, and by Corollary 6.6 it is not fixed by a's subjective triple. The only supplier of an objective value the framework's definitions provide is Definition 5.12, which requires an external *witness*; that requirement, and the open boundary around it, are recorded in Section 7.2. These last sentences *consume* Corollary 6.6 and Lemma 5.27 and re-prove neither. The answer is therefore four-way, over all 48 pairs: two transmutations admissible from every active source; six admissible from four sources each; six exiting pairs, whose target is the absolute exterior; and six (source, transmutation) pairs whose target is R_3^a and which no step of a carries out.

8.4. **Queries.** The blocked class shows that some region movements are outside an agent's reach. What an agent can do about them is ask – and asking is itself an artifact, classified by the same machinery.

Definition 8.21 (Query)**.** A *query* of an agent a is a pair (q, v) where $q \in \mathcal{A}$ is an artifact that denotes a subject $x = \text{denotes}_a(q)$ (Definition 5.30) and $v \in \{0,1\}^3$ is a designated *target cell* with $v \neq \text{Region}_a(x)$, such that the step $x\colon \text{Region}_a(x) \rightsquigarrow_a v$ is not realizable by a (Definition 8.12). The *requested transmutation* of the query is

$$g_{(q,v)} \;=\; \text{Region}_a(x) \oplus v \;\in\; \mathbb{Z}_2^3 \setminus \{(0,0,0)\},$$

the unique group element carrying the subject from where it stands to where the query asks it to stand (Proposition 4.20(i)). A question is thus typed by the transmutation it requests; this typing of queries by group elements is unrelated to the qualitative type $\tau(a)$ of an agent (Definition 7.1), which is a subset of region indices.

Theorem 8.22 (Query horizon)**.** *Let (q, v) be a query of a (Definition 8.21). The following hold:*

(i) $q \in \Theta_a$ *– every query of a is itself a symbol for a;*
(ii) *the subject x lies in a's classified domain $\mathcal{A}_a^+$, within the active regions; in particular $x \in \text{KNOWN}_a$;*
(iii) *an agent with $\Theta_a = \emptyset$ poses no queries at all, and no agent can pose a query whose subject lies in its own unknown U_a.*

Proof. (i) By Definition 8.21 the artifact q denotes, so q lies in the domain of denotes_a, which is Θ_a by the domain-restriction clause of Definition 5.30: only Θ_a-artifacts denote for a.

(ii) By the target clause of the same definition, every denotation target lies in $\mathcal{A}_a^+$ within a's active regions. Each of the six active cells has $M_a = 1$ or $C_a = 1$, so $x \in \text{KNOWN}_a$ (Definition 5.8).

(iii) If $\Theta_a = \emptyset$, part (i) leaves no candidate for q, so such an agent is query-free as well as symbol-free. By Corollary 7.5 the condition is exactly $R_4^a = R_7^a = \emptyset$. The count attached to it depends on which reading of K is in force, and the two figures must not be run together. Under the K-objective reading, in which Theorem 7.2 and Corollary 7.5 are stated and under which R_3^a may be inhabited, exactly 17 of the 109 labelled types satisfy it; this clause of the theorem quotes that count and is to that extent read outside the convention of Remark 8.11. Under the convention itself the acting agent has $R_3^a = \emptyset$, so only the 52 types with $3 \notin \tau(a)$ are available to it (Section 7.2), and of those exactly 8 are symbol-free: the criterion of Theorem 7.2 restricted to $S \subseteq \{1,2,5,6\}$ reads $S \cap \{1,5\} \neq \emptyset$, $S \cap \{2,6\} \neq \emptyset$ and $S \cap \{5,6\} \neq \emptyset$, whose failures are $S \subseteq \{2,6\}$ (4 sets), $S \subseteq \{1,5\}$ (4) and $S \subseteq \{1,2\}$ (4), with pairwise intersections of sizes 1, 2 and 2 and triple intersection of size 1, giving $16 - (12 - 5 + 1) = 8$. Neither count is asserted of the other reading. For the second clause, U_a is the complement of KNOWN_a (Definition 5.26), which part (ii) excludes. □

Three consequences are worth separating. First, elicitation ceases to be a metaphor: a query is an artifact with a cell, a subject and a group-element type, and Theorem 8.22 constrains all three. Second, the symbol-free types of Corollary 7.5 acquire an operational reading they previously lacked – an agent without symbols is not merely one whose denotation function has empty domain, but one that cannot pose a question. Third, part (iii) upgrades one clause of Remark 5.16 from conjecture to theorem: the clause that names the reservoir *un-elicited*, i.e. that what lies outside a's semantic body cannot be the subject of a's own queries, is now proven. The remainder of that remark – that the reservoir holds latent static content beyond the functional slice R_3^a, and that the region a latent element enters is fixed by its content – is untouched and stands as conjectured.

Corollary 8.23 (Symbol-free types are query-free under every architecture)**.** *Let a be an agent with $4 \notin \tau(a)$ and $7 \notin \tau(a)$. Then a has no query in the sense of Definition 8.21, and this holds whatever architecture α_a the agent carries.*

Proof. By Corollary 7.5 the hypothesis gives $\Theta_a = \emptyset$, so $\mathrm{denotes}_a$ has empty domain (Definition 5.30). A query of a is a pair (q, v) whose first component denotes for a (Definition 8.21), and no artifact does; the pair therefore fails at its first clause. The remaining clauses – the target cell and the non-realizability of the corresponding step – are the ones on which α_a bears, and they are never reached, so the conclusion is independent of α_a. □

Remark 8.24 (Where the content of this chain actually lies). It is worth being exact about how much each step above carries, because the chain is short and its steps are of very unequal weight. Clause (i) of Theorem 8.22 unfolds two definitions and stops: a query's first component denotes, the domain of $\mathrm{denotes}_a$ is Θ_a, hence that component lies in Θ_a. Nothing is established there that was not already fixed when denotation was restricted to the overlap. The first half of clause (iii) is the contrapositive and is equally immediate. The criterion of Corollary 7.5 is likewise a matter of unfolding, since $\Theta = M \cap C$ is by construction the union of the two cells on which M and C both hold, so its emptiness for a is exactly the absence of those two cells from $\tau(a)$. We do not present any of these as substantive theorems.

Two things in the chain are not definitional. The first is the count: that exactly 17 of the 109 types satisfy the criterion is a finite computation, and in the language of Remark 7.3 it is the number of covers of $\{M, C, K\}$ in which no block carries both M and C. The second is the quantifier in Corollary 8.23, and it is the one that repays attention: the obstruction holds for every architecture the agent might carry, and it does so for a structural reason rather than by inspection of the possibilities. A query has clauses of two kinds – one about the denoting token, the others about the target cell and the non-realizability of the corresponding step – and the architecture bears only on the second kind. An agent failing the first clause fails before any architecture-sensitive clause is reached, so no architecture can repair the failure and none needs to be examined.

The contribution here is therefore the composition rather than any of its links. A datum recording only which cells an agent occupies – no artifacts, no operations, no denotations – suffices to settle an operational question about that agent under every architecture whatsoever. That a bound this coarse survives quantification over architectures is what the section is for; the individual implications that carry it are, as noted, small.

Remark 8.25 (The direction of the implication, and what the count does not say). Two boundaries, both of which the statements above respect and neither of which they exceed. First, the implication runs one way. A query of a requires $\Theta_a \neq \emptyset$ by Theorem 8.22(i), and nothing here asserts the converse: a non-empty carrier does not by itself supply a query, since Definition 8.21 asks in addition for a step that a cannot realize, and realizability is read off a region assignment (Definition 8.12) rather than off $\tau(a)$. Second, the seventeen types of Corollary 7.5 are those that are query-free *for want of a carrier*. An agent whose carrier is non-empty may still admit no query, under its architecture and at its assignment, and this paper settles no count for that case. The phrasing matters, because whether an agent has a query is not in general a property of its type: by Definition 8.21 it turns on $\mathrm{denotes}_a$, which Definition 5.30 leaves as a datum of the agent, and on which steps are realizable, which Definition 8.12 reads off an assignment. The seventeen are accordingly the types that force query-freeness under *every* such completion, and not a count of the agents that happen to have none.

Remark 8.26 (The general query). A general query – "what am I missing?" – takes the agent's own self-model as its subject. By Remark 6.7 that self-model is conjectured to lie in R_7^a, and by Corollary 6.6 the population of R_3^a is not fixed by a's predicate triple under the subjective reading of K. Nothing more is claimed here: the conjectural status is inherited from Remark 6.7, and no result below depends on this remark.

8.5. **Three Memories.**

Definition 8.27 (The three memories of an agent)**.** For an agent a with architecture α_a (Definition 8.12), define

$$\begin{aligned}
\mathrm{Sto}(a) &= (M_a,\, C_a,\, K_a),\\
\mathrm{Exe}(a) &= \{\, (f,h) \in K_a \times K_a \mid f \circ_a h \text{ is defined} \,\},\\
\mathrm{Lnk}(a) &= \big(\mathrm{denotes}_a \restriction \Theta_a,\ \{\, ((f,h),\, f \circ_a h) \mid (f,h) \in \mathrm{Exe}(a) \,\}\big),
\end{aligned}$$

called respectively a's *storage*, *execution* and *linkage* memory: storage holds the artifact extensions; execution holds what a GEN step consumes; linkage holds the record of denotation on Θ_a together with the derivation relation of α_a – which pair produced which artifact. This makes precise, and is the single definition point for, the three-memory reading recorded in Remark 7.7; linkage carries there the same two components as here, denotation and provenance, so that the word has one extension in both places.

Remark 8.28 (What "memory" does and does not import). The word names the three structures just defined and nothing further. $\mathrm{Sto}(a)$ is a triple of subsets of $\mathcal{A}$, $\mathrm{Exe}(a)$ a set of pairs on which $\circ_a$ is defined, and $\mathrm{Lnk}(a)$ a pair consisting of a function restriction and a relation; each is fixed by the agent's triple together with its architecture, and none carries a time parameter. The step relation of Definition 8.12 has no time index, so nothing here encodes persistence, retention, update over time, forgetting, or capacity, and no result below depends on reading the three structures as a model of memory in any of those senses. Where the word is used, it abbreviates "the storage, execution and linkage structure of Definition 8.27".

Corollary 8.29 (Linkage is not determined by storage)**.** $\mathrm{Lnk}(a)$ *is not a function of* $\mathrm{Sto}(a)$*: there exist agents with identical storage memory and distinct linkage memory. Consequently an agent holding only its artifact extensions cannot recover the provenance of its own artifacts.*

Proof. Take any K_a with $|K_a| \geq 2$. By Lemma 8.7 there are architectures $\alpha_1 \neq \alpha_2$ with $K_{\alpha_1} = K_{\alpha_2} = K_a$ and $d_{\alpha_1} \neq d_{\alpha_2}$; in the witness there constructed, $\circ_1$ is nowhere defined while $\circ_2$ satisfies $a_1 \circ_2 a_1 = a_n$. Two agents carrying these two architectures over the same (M_a, C_a, K_a) have the same Sto by construction, while their derivation relations differ: the first is empty, the second is not. Hence Lnk differs. □

The denotation component separates for a second, independent reason: $\mathrm{denotes}_a$ is a datum of the agent not determined by its triple, and Definition 5.30 is explicit that two agents sharing a symbol may send it to different referents. Both components of linkage are therefore invisible in storage, which settles affirmatively the question left open at Remark 7.7 whether the three memories are genuinely three.

The same witness settles a second question, and it is the one that fixes the scope of the type classification. An agent's *type* $\tau(a)$ (Definition 7.1) is read off its triple; its architecture

is a further datum. The next statement records that the second is not recoverable from the first.

Theorem 8.30 (Type–architecture separation)**.** *Let $M, C, K \subseteq \mathcal{A}$ with $|K| \geq 2$. There are agents a and b with $(M_a, C_a, K_a) = (M_b, C_b, K_b) = (M, C, K)$ – hence with equal types, $\tau(a) = \tau(b)$ – carrying architectures $\alpha_a \neq \alpha_b$ for which*

$$\mathrm{Exe}(a) \neq \mathrm{Exe}(b) \qquad \textit{and} \qquad \mathrm{Lnk}(a) \neq \mathrm{Lnk}(b) \qquad (\textit{Definition } 8.27).$$

Neither Exe *nor* Lnk *is therefore a function of τ.*

Proof. Apply Lemma 8.7 to (M, C, K): it yields $\alpha_1 \neq \alpha_2$ over that same triple, each with basis contained in K, so each is an agent architecture (Definition 8.12). Let a carry α_1 and b carry α_2. Equality of the triples gives $\tau(a) = \tau(b)$, the type being the set of regions the triple inhabits. In α_1 the composition is nowhere defined, so $\mathrm{Exe}(a) = \emptyset$; in α_2 the composite $a_1 \circ_2 a_1$ is defined with $a_1 \in K$, so $(a_1, a_1) \in \mathrm{Exe}(b)$. Hence $\mathrm{Exe}(a) \neq \mathrm{Exe}(b)$. The derivation component of Lnk is indexed by Exe, so it is empty for a and non-empty for b, giving $\mathrm{Lnk}(a) \neq \mathrm{Lnk}(b)$. □

Remark 8.31 (What the separation leaves open)*.* The theorem separates Exe and Lnk, each of which is fixed by the triple together with $\circ_a$. It says nothing about the availability of queries. Whether a given step is realizable is read off a region assignment and not off the architecture alone (Definition 8.12), so query availability depends on more than the pair (τ, α) and is not classified here.

The corollary also locates precisely what is missing. Every $x \in K_{\alpha_a}$ of positive depth is reached from the basis by finitely many applications of $\circ_a$ (Definitions 8.2 and 8.3), so a derivation for it exists; nothing in that fact makes the agent *hold* one. The gap between the existence of the chain and its being recorded is exactly linkage memory. This is why the third memory is the non-obvious one: storage and execution are visible in the extensional theory – they are the extensions and the composition table – whereas linkage is provably invisible to it (Corollary 8.29), and it is precisely what any account appealing to derivation order presupposes without recording.

8.6. **The Closure of Internal Execution.**

Scope and finiteness boundary. The theorem below, like the stabilization results of Definition 8.3, is asserted over *finite* artifact domains only, where the generation sequence of Definition 8.2 terminates in finitely many steps. It is not extended, silently or otherwise, to the infinite case; the classification results of Sections 3 and 4 it uses are domain-general.

Theorem 8.32 (Fixed point of internal generation)**.** *Let $\mathcal{A}$ be finite and let a be an agent with architecture $\alpha_a = (\mathcal{O}_a, \circ_a)$ (Definition 8.12). Call an artifact* obtainable *from a given assignment if some finite run of* GEN *steps of a from it makes the artifact functional for a. The following hold:*

(a) *from an assignment whose functional extension is the basis, $K_a = \mathcal{O}_a$, the set of artifacts obtainable by a, together with $\mathcal{O}_a$ itself, is exactly the compositional closure $K_{\alpha_a} = \mathrm{cl}_{\circ_a}(\mathcal{O}_a)$;*

(b) *the associated operator $\Phi(X) = X \cup \{f \circ_a h \mid f, h \in X,\ f \circ_a h \text{ defined}\}$ is monotone on the finite lattice of subsets of $\mathcal{A}$, and the iteration $\Phi^n(\mathcal{O}_a)$ attains its least fixed point above $\mathcal{O}_a$ after exactly $n^*(\alpha_a)$ iterations and not fewer;*

(c) *an assignment with $K_a = K_{\alpha_a}$ is a fixed point of internal generation: no* GEN *step of a is realizable from it at all, so no run of* GEN *steps extends K_a beyond K_{α_a}.*

Proof. (a) By Definition 8.12 a GEN step reads its operands off the current functional extension and adds its output to it, so a run of such steps produces extensions $X_0 = \mathcal{O}_a$, $X_{k+1} = X_k \cup \{f \circ_a h\}$ with $f, h \in X_k$. Now K_{α_a} is closed under $\circ_a$: if $f, h \in K_{\alpha_a}$ then both lie in some $K^{(n)}_{\alpha_a}$ and $f \circ_a h \in K^{(n+1)}_{\alpha_a} \subseteq K_{\alpha_a}$ (Definition 8.2). Hence $X_k \subseteq K_{\alpha_a}$ for every k, the reachable set lying in the closure. For the converse, induct on depth (Definition 8.3): elements of depth 0 are $\mathcal{O}_a$, and an element x of depth $n+1$ equals $f \circ_a h$ for some $f, h \in K^{(n)}_{\alpha_a}$, each obtainable by the inductive hypothesis. Concatenate the run obtaining f with the run obtaining h, deleting from the second every step whose output is already functional when that step is reached. Both preconditions of a GEN step (Definition 8.12) then hold at every surviving step, and this is what the deletion is for: the operand condition $f, h \in K_a$ is monotone in K_a and so survives the enlarged starting extension, whereas the condition $K_a(y) = 0$ is anti-monotone and can fail. The deletion is well defined and loses nothing, because K_a only grows along a run: an induction on the second run shows that the extension after its k-th step is contained in the extension reached at the corresponding point of the concatenation, a deleted step's output already lying there, so each surviving step still finds its operands and each deleted step's output is functional anyway. The concatenated run therefore makes both f and h functional, and after it either x is functional already or one further GEN step obtains it.

(b) Φ is monotone and $\Phi^n(\mathcal{O}_a) = K^{(n)}_{\alpha_a}$ by Eqs. (8.1) and (8.2). The least fixed point above $\mathcal{O}_a$ is K_{α_a}, and $n^*(\alpha_a)$ is by Definition 8.3 the *minimum* n with $K^{(n+1)}_{\alpha_a} = K^{(n)}_{\alpha_a}$, whence "exactly" and "not fewer". Finiteness of $\mathcal{A}$ guarantees the minimum exists.

(c) A GEN step requires $f, h \in K_a$ and $K_a(y) = 0$ for $y = f \circ_a h$ (Definition 8.12). With $K_a = K_{\alpha_a}$ the closure property just used gives $y \in K_{\alpha_a} = K_a$, so $K_a(y) = 1$ and the precondition fails for every admissible pair. By (a) this assignment is the one reached from $\mathcal{O}_a$, and $\mathcal{O}_a = \mathcal{F}^{(0)}_{\alpha_a}$ is the depth-zero class (Definition 8.3), which no artifact of positive depth joins. □

The consequence is the point of the section, and its boundary is the boundary stated after Definition 8.12. For a fixed basis $\mathcal{O}_a$ and a fixed composition $\circ_a$, the functional set a can reach is exactly K_{α_a}, and by (c) no GEN step extends it. That is a statement about a fixed architecture, and the architecture is the parameter Definition 8.12 holds fixed: the relation offers no step that changes $\mathcal{O}_a$ or $\circ_a$, and none by which an artifact enters a's classified domain from outside it. So the theorem says where internal generation stops, and says nothing about what would carry an agent past that point, whether anything can, or where it would come from. Were the basis enlarged, the resulting functional set would again be a compositional closure – the operator $\mathrm{cl}_{\circ_a}$ of Definition 8.5 applied to the larger basis, an identification with an existing operator rather than a new one – but such an enlargement is a change of the fixed parameter, not a step of the relation, and nothing here establishes that it occurs. A query (Definition 8.21) is likewise not a step: it is an artifact typed by a transmutation its author cannot carry out, whose subject is one a already holds (Theorem 8.22(ii)). Asking, answering, and being given are all outside the relation, and the execution layer settles only what falls inside it.

Remark 8.33 (What this section does not claim). No cost, no trajectory, no adjacency topology on the regions and no graded predicate is introduced above; the step relation carries

no metric and no time index, and $\rightsquigarrow_a^*$ is a reflexive-transitive closure rather than a temporal process; the finite sequences of generation steps in Theorem 8.32 are derivation orders in the sense of Definition 8.3, not runs. Those quantitative questions – how far a movement carries, at what cost, and along which route – are open; the verifier transitions among them are treated in the companion on region transitions [Bra26], and the cost and route are treated nowhere, on the division of labour recorded in Section 10. Nor is acquisition claimed or modelled: the relation holds the agent architecture fixed and has no supply operation. What is settled here is the coherence question: which (source, transmutation) pairs are admissible, where an agent's own outputs land, what a question is, and where internal generation stops.

9. Positioning

9.1. **Related Work.** The MCK framework assembles familiar ingredients – approximation regions, agent-indexed epistemic status, unawareness, attribute lattices, generated structure – into a specific configuration. To position the contribution honestly, this subsection compares the framework with the neighbour families below, stating in each case what is shared and where the framework departs. The comparison covers every layer of the paper, the execution layer of Section 8 included; where a claimed departure does not survive scrutiny it is withdrawn here rather than defended.

Rough set theory. Rough set theory [Paw82, Paw91] is the closest formal relative of the region layer: it also classifies a domain, relative to a classifier's information, into a positively characterized part, a boundary of uncertain status, and an exterior. The structural difference lies in how the regions arise. In rough set theory the classification is *derived*: an indiscernibility equivalence over attribute information induces lower and upper approximations, and the boundary is their difference. In MCK the regions are *generated* by three independent primitive predicates (Definitions 3.2 to 3.4): Theorem 4.8 partitions the classified domain into $2^3 - 1 = 7$ cells rather than three, and no indiscernibility relation is assumed.

We withdraw a stronger claim that would not survive scrutiny. A Pawlak information system takes a finite non-empty attribute set, each attribute a total single-valued function on the objects; taking $\{M, C, K\}$ as the attributes makes the indiscernibility classes exactly the seven regions together with the exterior, so the distinction between an opaque inert artifact (R_2, Definition 4.2) and an opaque operator (R_6, Definition 4.6) is perfectly representable there. The difference is not expressive range but what is taken as given: rough set theory *derives* an approximation from an indiscernibility relation relative to whatever attributes are supplied, and studies that derivation in general; MCK *stipulates* three attributes with a fixed reading and studies the cell structure they determine. The same correction applies to the three-way decision refinement of rough sets [Yao10], which graduates the *certainty* of a classification and is orthogonal to the functionality axis rather than short of it. One structural difference does survive. The semantic pair (M, C) is stipulated non-exclusive (Axiom 3.11) and its overlap carries the symbol apparatus (Definition 4.16), so it is not an orthopair in the powerset setting of the algebraic treatments of rough approximation [CC18, Eq. 22, p. 576], where orthogonality of the positive and negative parts is disjointness.

Epistemic logic. The agent-indexed layer of Section 5 has an evident ancestor in epistemic logic [Hin62, FHMV95]: relativized regions (Definition 5.6) and per-agent known domains play a role analogous to agent-indexed knowledge operators, and region non-invariance (Lemma 5.20) is a set-theoretic analogue of the observation that different agents know different things. The departure is methodological. MCK is *extensional* and *static*: there are no possible worlds, no accessibility relations, and no closure of knowledge under logical consequence – the logical-omniscience problem does not arise because no propositional attitudes are modeled. The framework classifies *artifacts*, not propositions, and its multi-agent layer is a parametric instantiation of the same predicate triple (Definition 5.6) rather than a modal operator that can be nested or iterated.

Ability, knowing how, and affordances. The functionality predicate K (Definition 3.4) has older and closer neighbours, and they are worth naming precisely. That what an agent can do is distinct from what truths it knows is Ryle's knowing-how/knowing-that distinction [Ryl49, Ryl46], and it has since been given formal semantics of its own: Wang takes a knowing-how construct as primitive rather than compounding a knowledge operator with an action operator [Wan15], and alternating-time temporal logic carries an agent-indexed ability modality in an object language with no epistemic operator [AHK02]. Wang reads his construct in a particular sense – having a plan that never fails – so the alignment with K drawn below is ours rather than his. MCK stipulates the independence of K from M as an axiom (Axiom 3.12) and studies the cell structure it determines.

Two features are specific, and they are what the positioning claims. First, the axis is predicated of the *artifacts* and not of the agents: $K(x)$ says that x has operator-like character, where an ability modality says that an agent can bring something about. Second, it is extensional and simultaneous with the semantic pair: M, C and K are evaluated on one domain at one semantic state, so their independence yields a partition into cells (Theorem 4.8) rather than a family of formulas, and questions about which combinations an agent realizes become combinatorial (Theorem 7.2). Neither feature is offered as an increase in expressive power over the modal frameworks; the difference is in what is made primitive and what is then proved about it.

The concession extends to the recovered cell. Gibson's *affordance* [Gib79] is functionality determined jointly by an artifact and an agent, which is close to the objective/subjective split on K (Definition 5.14) and to functional potential (Definition 5.12); the difference is that here the distinction is carried as a cell of a proven partition and is reactivated per agent by an explicit criterion (Proposition 5.15), not that the underlying relation is new. An agent's own unresolved capacities sit at R_6 (Definition 4.6), registered in its ontology without being resolved – which is what $C = 1$ records, while $K = 1$ records that they operate. The framework places that familiar configuration rather than explaining it.

Awareness and unawareness logics. For the treatment of UNKNOWN and of the absolute exterior – artifacts *not even recognized* – the strongest neighbours are awareness logic [FH87], the partitional unawareness structures of Modica and Rustichini [MR99], and the interactive-unawareness structures of Heifetz, Meier, and Schipper [HMS06]. The shared concern is precisely what lies beyond an agent's conceptual grasp: the semantic horizon (Definition 6.1) and the per-agent unknown U_a play a role analogous to the complement of an awareness set, and the semantic collapse of Remark 5.28 – an agent cannot, from

inside its own state and under the subjective reading of K, distinguish the two strata of its unknown – echoes the introspective limits that motivate unawareness structures.

The departure needs stating with care, because its most visible feature – that the framework classifies *artifacts* rather than propositions – is itself an established line and not a departure at all. Object-based unawareness models an agent's failure to conceive of elements of the domain rather than its failure to entertain a proposition, and the relation between that approach and the subjective-state-space approaches above is worked out by Board, Chung, and Schipper [BCS11]. Unawareness over objects is established there.

What MCK adds is a stratification of the unaware domain by objective functionality. By Proposition 3.8 the unknown splits into the functional-yet-unrecognized cell R_3 and the absolute exterior, and the K-objective reading recovers the former per agent (Remark 5.19, Proposition 5.15) as the region of functional potential (Definition 5.12). MCK therefore carries a second, functional attribute on the unaware part of the domain, so that an object operative within an agent's operational scope while lying outside its conceptual reach has a cell of its own. In the model of Board, Chung and Schipper [BCS11] awareness of an agent at a state is a set of objects, with no second attribute carried on the objects it excludes; the difference lies in what is made primitive and what is then proved about it. Among the awareness and unawareness formalisms we have examined directly, we have not found the unaware domain carved by objective functionality in this way – a statement about our coverage, not about the literature.

The query obstruction of Corollary 8.23 invites a comparison with this family too, and the comparison is worth making carefully, because the objects turn out to sit in different places. In the formalisms above, the material from which awareness is built is supplied by the modeller – a language, a set of atoms, a domain of objects, an index set of questions – and awareness is a filter on it: in the object-based setting of Board and Chung [BC21, §2.3, p. 14], for instance, awareness is a correspondence assigning to each world the set of objects the agent is aware of at it. In MCK the corresponding material is not supplied alongside the agent but is part of what the agent is classified as holding: a query's token is an artifact in the agent's own domain, carrying a cell of the same partition as everything else it holds (Definition 8.21). The obstruction proved here is about that token's availability, and it is for that reason that a purely occupancy-level datum can settle it.

The distinction is sharp enough to be tested rather than asserted, and Board and Chung's axiom system supplies the test. Their axiom A1 makes $A_i\alpha$ an axiom for every formula α with no free variables [BC21, §2.2, p. 13], so an agent aware of no object whatever is nonetheless aware of every closed formula – which looks, at first reading, like an agent with an empty carrier that retains something. Their own soundness argument shows what the awareness consists in: the semantic clause for A_i requires $V(x) \in \mathcal{A}(w)$ for every x free in α, and when α has no free variables that requirement is met vacuously [BC21, p. 27], whatever $\mathcal{A}(w)$ happens to be. No object is possessed and none is needed. Nothing in this instantiates Definition 8.21, which asks for an artifact q in the agent's classified domain, a subject $x = \mathrm{denotes}_a(q)$, a target cell, and a step that a cannot realize; a formula true at a world by vacuous quantification supplies none of the four. The two statements concern different formal objects: neither entails the other and neither contradicts it. The comparison is nonetheless informative, and that is why it is drawn here – it locates precisely where the two frameworks put the material a question is made of.

One expressive resource in this literature does something MCK cannot, and it is worth naming rather than passing over. Halpern and Rêgo extend the language with quantification

over propositional variables, whose values are quantifier-free formulas, so that an agent can be said to know $\exists x \neg A_i x$ – that there is a formula it is unaware of – without any particular formula being named [HR06, §3, pp. 4–6]. MCK has no quantified syntax of that kind: its queries are typed by group elements (Definition 8.21), and a question about an unnamed subject has no representation among them. Theorem 8.22(ii) accordingly confines a query's subject to the agent's known domain, and that is a boundary of what the present language expresses rather than a claim that the operation is impossible. Extending the query apparatus to quantified subjects is an open direction, and the resource just described is the natural model for it.

Finally, one methodological point in this family bears directly on the modelling choice made in Axiom 3.11 and Definition 4.17. Dekel, Lipman and Rustichini, having shown that standard state-space models preclude unawareness, raise in passing the very reading on which the present framework rests – whether a statement an agent has not thought of is "simply not in his internal language" – and record that their argument does not require them to take a position on it [DLR98, p. 169]. The present framework does not settle that question either, and no reduction between the two settings is offered here. What it does is take an explicit stance on a related boundary of its own: denotation is confined to Θ (Definition 4.17), so within this framework an artifact an agent has no token for is outside the only internal language the formalism defines. That is a modelling choice, recorded as one; its warrant is that it is explicit and that its consequences are drawn.

Belnap's four-valued logic. The joint presence or absence of M and C yields four cases that superficially echo the four values of Belnap's useful four-valued logic [Bel77] – told true, told false, told both, told neither. The analogy stops at the count, and it stops for a structural reason rather than for the alethic one an earlier version of this section gave. Belnap's values are epistemic rather than ontic in his own statement of them: an answer, he writes, "does not have the ontological force, 'That's the way the world is,' but rather the epistemic force, 'That's what I've been told . . . '", and a set-up records not what is true but what the computer has been told. His four values are accordingly the four subsets of a *single* register – the two polarities, told-true and told-false, of one question about one proposition – so "told both" is a conflict of reports and not a contradiction in the world. The four M/C cases have a different shape. M and C are two *distinct* registers, integration and residual opacity, neither of which is the negation or the polarity-reverse of the other, and $\Theta = M \cap C$ (Definition 4.16) is their overlap – *partial integration*, structured meaning coexisting with residual opacity, the natural habitat of symbols and of the denotation function (Definition 4.17) – rather than a doubled answer to one question. We state the consequence explicitly: the framework does not read $M \cap C$ as contradiction and makes no paraconsistency claim; the predicate layer is entirely classical.

Formal Concept Analysis. Formal Concept Analysis [GW99] is a closer relative of the region layer than a passing mention would suggest, and the relationship is worth conceding precisely rather than deflecting. Take the formal context whose objects are the artifacts of $\mathcal{A}$, whose attributes are M, C, K, and whose incidence is predicate satisfaction. The object intents are then exactly the predicate triples, and the seven regions (Definitions 4.1 to 4.7) are the non-empty *fibres* of the object-intent map – sets of objects, one for each realized non-empty intent – so that the poset of Section 4.6 is the order those intents inherit from inclusion of attribute sets. One clarification keeps the concession accurate: that poset is

not the concept lattice of the context. A concept lattice is complete, whereas Remark 4.19 establishes that the region poset has no universal meet, the meet of two atomic regions being the empty attribute set, which the classified domain excludes. Subject to that, fixing an intended reading for three attributes is a modeling choice, not a strengthening of the formalism. Nor is the agent relativization a departure – a family of contexts over a shared object set is still FCA data.

Two layers contain material that lies outside a single formal context, and they are the honest answer to the comparison. The compositional closure of Section 8 (Definition 8.2) is a closure under a partial binary operation on the extension of one attribute – data alongside the three components a context consists of, rather than the Galois closure of an incidence relation. The execution layer of Section 8 takes region *assignments* as its objects and *steps between them* as its relation (Definition 8.12), so Theorems 8.16 and 8.32 are statements about a relation between incidences, where a context carries one. The observation concerns the three components as defined, and is compatible with whatever a context may be used to encode by other means. Theorem 8.17 is not in that class: it is a statement about the setwise stabiliser of a six-element subset of the cube, hence about attribute triples, which are formal-context data. The predicate-toggle group (Proposition 4.20) is likewise a structure on the attribute set – an involution per attribute – rather than on the concept lattice.

Enumerative combinatorics of covers. The classification of Theorem 7.2 has a neighbourhood that is neither semantic nor logical but combinatorial, and it is the closest of all, because the objects coincide. Reading a region as the set of predicates that hold on it turns an agent's type into a family of non-empty subsets of $\{M, C, K\}$, and the requirement that the three extensions be non-empty into the requirement that the family cover $\{M, C, K\}$; the admissible types are therefore the covers of a three-element set, and the two counts of that theorem are the labelled and unlabelled cover numbers, in print since at least the early 1970s (Remark 7.3) [Com74, BLL98]. The derivation this paper gives is the standard one specialised to $n = 3$ – inclusion–exclusion, then Burnside – so nothing in the arithmetic is new here, and we do not present it as such. What the present setting adds is the reading under which the object is worth counting: the family is an agent's occupancy pattern, the ground set is the agent's own predicate set, and the questions asked of the covers afterwards – which of them bear a symbol, which force the absence of a query, which structural profiles they realize – are questions about agents rather than about set systems. That the same finite objects carry both readings is what makes the classical count usable here, and saying so is more useful than a claim of novelty that a combinatorially literate reader would correct.

The identification also makes it possible to ask which of this paper's results are already fixed by the combinatorics and which need the reading of the three predicates, and the answer separates them cleanly. The set of admissible types is symmetric in M, C and K: relabelling the three predicates permutes the covers and fixes the family, which is why the two counts are the classical ones and why the orbit count of Theorem 7.2 is taken under exactly that action. Symbol-freeness is not symmetric in the same way. The property singles out the blocks containing both M and C, and a relabelling can carry a symbol-free type to one that is not: $\tau = \{1, 6\}$ has empty carrier, while exchanging the roles of the predicates turns it into $\tau = \{3, 4\}$, which does not. What the modelling choice of Definition 4.16 contributes is therefore the selection of one pair from three, and with it the identity of the seventeen types of Corollary 7.5 – though not their number, since the same count is obtained whichever pair is distinguished. The combinatorial substrate is common property; the semantic content of

the later results lies in which cells are read as carrying symbols, and hence in which covers the framework picks out, rather than in how many there are.

Classifications, infomorphisms, and channels. The same data structure supports a second comparison, closer still, which we state explicitly rather than leave for a reader to discover. A *classification* in the sense of Barwise and Seligman [BS97] is a set of tokens, a set of types, and a support relation between them; an MCK system is one, with the artifact domain as tokens, $\{M, C, K\}$ as types, and predicate satisfaction as support. The correspondence goes further than the structure. The regions are the type sets realized by tokens; the framework's stipulations have the shape of entailment constraints on that classification – the R_3 suspension (Axiom 4.11) is the constraint $K \vdash M, C$, every token supporting K supporting M or C, and M/C non-exclusivity (Axiom 3.11) is the refusal of the constraint that would forbid a token from supporting both; and the multi-agent layer of Section 5 is a family of classifications over a shared token set, which is precisely the setting in which infomorphisms and information channels are defined.

What this paper adds is a fixed choice and its consequences: three types under a stipulated reading – a semantic pair whose overlap carries its own apparatus (Definitions 4.16 and 4.17) and an independent functionality axis – together with the combinatorics that choice determines (Theorems 4.8 and 7.2) and the layers built over it. Barwise-Seligman theory is a general account of classifications and their morphisms; this paper studies one three-type classification under a fixed interpretation.

The comparison also locates a genuine gap. The framework builds an external witness into functional potential (Definition 5.12 and Lemma 5.22), and it has no resource for saying when one agent's certification transfers to another, or what to do when two witnesses disagree; channels and local logics are the natural machinery there. We record the question as open below rather than as answered.

Epistemic opacity of computation. The philosophical literature on the epistemic opacity of computational processes [Hum09] and on transparency in complex computational systems [Cre20] is the *motivational* neighbour of the predicate C and of the opaque-operator region R_6 (Definition 4.6): artifacts that are used, and demonstrably work, without being understood. That literature analyzes opacity and transparency as epistemological properties, in philosophical terms and at several levels of granularity. MCK's complement is formal: it gives the opaque-but-functional condition a definite cell in a proven partition (Theorem 4.8) and to make it agent-relative: the same artifact can be a transparent operator for one agent and an opaque operator for another (Lemma 5.20).

Symbols and grounding. Harnad's symbol grounding problem [Har90] asks how the tokens of a symbol system acquire meaning otherwise than by reference to further tokens that are themselves meaningless to the system manipulating them. MCK does not answer that question and must not be read as answering it: denotes is stipulated, its codomain Ref is primitive, and grounding is therefore presupposed rather than explained. What the framework contributes is a placement, and one that follows from the definitions. Because $C(x) = 1$ records a *residual* incompleteness rather than the total absence of a certificate (Definition 3.3), an artifact with $M = 1$ and $C = 0$ is fully resolved and retains no unresolved remainder with which to point beyond itself; symbolic character therefore requires $C = 1$ alongside $M = 1$, which is the cell Θ. The argument settles where symbols can live, not how they come to mean.

The opaque-operator cell R_6 (Definition 4.6) invites comparison with Searle's Chinese Room [Sea80], and the comparison has to be drawn carefully, because the occupant's own report does not supply the predicates. Searle has his occupant say that he is "not even confident that I could recognize Chinese writing as Chinese writing", and the argument's closing pages deny that the manipulations are symbol manipulations at all, "since the symbols don't symbolize anything". What the text does supply is individuation by shape: the occupant can identify the symbols entirely by their shapes. Registration under a shape criterion is enough for $C = 1$ as Definition 3.3 defines it – apprehension with resolution incomplete, which does not require recognition under a semantic kind – while the rule-following gives $K = 1$ and the absence of a certificate gives $M = 0$. On that reading, which is ours and not Searle's, the occupant's state with respect to the batches is $(M, C, K) = (0, 1, 1)$. The framework's interest in the example is classificatory: the configuration is a cell of a proven partition (Theorem 4.8) and is agent-relative (Lemma 5.20). On the informational side, Floridi's analysis imposes a veridicality condition on semantic information in his sense [Flo05], while M is a record of certification: $M(x) = 1$ says that a certificate of understanding exists (Definition 3.2), and the framework's results are stated without a truth predicate.

Generated structure. The closure machinery of Section 8 has neighbours of its own. The compositional closure K_α (Definition 8.2) is an instance of generated-subalgebra machinery specialised to a single partial binary operation: it is the subalgebra of $(\mathcal{A}, \circ)$ generated by the operational basis [Bir35]. We build it by the bottom-up iteration of Definition 8.2, which computes the same set for a partial operation on a finite carrier. The phenomena the layer records are familiar in outline from clone theory, whose objects of study are sets of functions closed under composition and the lattice such closures form [CV08]: there too distinct bases may generate the same closure, which is the phenomenon Theorem 8.30 turns to account here.

What the closure is used for in this paper is narrower than what those literatures study. It supplies the architecture over which Theorem 8.32 locates the fixed point of internal generation, and the depth function by which Lemma 8.7 separates two architectures over one triple. Neither use requires the lattice of closures, and none is claimed beyond them.

The execution layer. The step relation of Section 8 (Definition 8.12) is deliberately thin: no time index, no dating of steps, no cost model. What is specific is what the states are and how the admissible moves are characterized. The state space is not a set of arbitrary labels but the region partition itself, so a step is a change of cell in a proven classification (Theorem 4.8) and the two kinds that carry a transmutation – movement and exit – are typed by a group element (Proposition 4.20 and Remark 8.13); and the moves coherent from every occupiable cell are identified as a setwise stabilizer (Theorem 8.17), which makes the answer a symmetry statement about the cell set rather than the outcome of a reachability computation.

Positioning. Tripartite approximation, agent indexing, unawareness in both its state-space and its object-based form, Boolean attribute lattices, generated subalgebras and the enumeration of covers are all established elsewhere, and the paragraphs above locate the framework against each. What this paper adds runs through them rather than beside them, and it rests on two modelling choices: that M and C may overlap (Axiom 3.11), and that denotation is restricted to their overlap Θ (Definitions 4.16 and 4.17). The results that follow are of unequal weight, and the paper says so where it proves them: the exact image of the structural

profile (Theorem 7.12) and the failure of the compositional closure to determine its own generation data (Corollary 8.8) carry the mathematical and the conceptual load, while the carrier condition on a query is definitional and what it contributes is the bridge from a finite type to an architecture-independent obstruction (Remark 8.24). Among the formalisms we have examined directly, we have not found that bridge drawn from the intersection of two independently specified predicates over artifacts; that is a statement about our coverage.

10. Limitations and Research Programme

Downstream companions. This paper is the static foundation. Two layers usually thought of as downstream are developed *in* it: the compositional layer on the functionality axis, a static-generative stratification of the functional set by derivation order (Definitions 8.1 to 8.3), and the execution layer (Section 8), a static step relation between region assignments carrying neither time index nor cost. Three further bodies of material are deliberately excluded rather than omitted. First, *region dynamics*: how artifacts move between regions over time – promotion, demotion, horizon crossing – is not treated here (Remarks 4.15, 4.21 and 6.3), and it remains open: the companion on region transitions [Bra26] establishes restricted realization mechanisms and their limits rather than movement along the transition operator, and supplies no promotion, demotion, or horizon-crossing formalism. The division of labour with that companion is worth stating precisely, because it is not a clean split between "static here" and "dynamic there". The present paper fixes *which* moves are coherent and *from which* cells: the admissible transition operators themselves (Proposition 4.20), the step relation over region assignments and the stabilizer characterization of the moves coherent from every occupiable cell (Definition 8.12 and Theorem 8.17), the constraints on which source and target cells they may connect (Remark 4.21), and the point at which the subjective reading of K leaves a cell undecided (Corollary 6.6). What is left open is the quantitative half: *how far* a move carries, *at what cost*, and *along what trajectory*. No cost model or trajectory is supplied here, and none is supplied by the companion work, which develops the denotation graph and its taxonomy of symbol kinds, and verifier transitions by which a token-less point is minted; it does not develop a metric or a topology, and nothing here assumes that it does. Second, a *cryptographic instance* of the transmutation calculus – keyed interpretation, per-agent region disagreement between key-holders and non-holders – is developed elsewhere. Third, *validator architecture*: the system-level question of how predicate evaluation is operationalized – who evaluates M, C and K on a given artifact, and under what discipline – is downstream work. These are forward pointers, not dependencies: no statement in this paper relies on the deferred material.

Limitations. Five limits should be stated plainly.

(1) *The certificate is not characterized.* $M(x) = 1$ is stated in terms of a certificate of understanding (Definition 3.2), and the paper constrains that notion only by the three conditions the definition names. An unrestricted reading, on which an artifact counts as certified merely because it can be displayed as the value of a known function on known inputs, makes M hold everywhere and the predicate separate nothing; the constraint that excludes such readings is not supplied here (Section 3.2). The predicate is also asymmetric in its evidence: a certificate exhibited settles $M(x) = 1$, whereas a failed search settles only that no certificate was produced by that search, so assignments of

$M(x) = 0$ – and the memberships of R_2, R_3 and R_6, whose defining conjunctions contain one – are relative to the search behind them. Both points concern the *selection* of an assignment rather than the results proved from one. Every theorem here is stated for an arbitrary assignment meeting its hypotheses, so its validity does not turn on which assignment is intended: what the limitation withholds is a criterion for choosing the intended one, not the consequences that follow once a choice is made.

(2) *Binary discretization.* The predicates take values in $\{0, 1\}$ (Definitions 3.2 to 3.4). Graded understanding, partial certificates, and degrees of opacity are representable only coarsely, through the structured ambiguity domain Θ (Definition 4.16). The boundary cases excluded by forcing binary values are an acknowledged excluded configuration of the discretization, flagged here in the same spirit as Remarks 3.9 and 4.14; their recovery through a graded refinement of the predicate layer is deferred to downstream work.

(3) *The R_3 suspension is a convention.* The suspension (Remark 4.12) is a justified framework convention, not a theorem. The multi-agent layer sharpens its status – under the K-subjective reading the emptiness of R_3^a becomes provable, and under the K-objective reading the cell reactivates (Proposition 5.15) – and the execution layer derives it for the artifacts an agent generates itself (Theorem 8.16), relative to the output-apprehension stipulation (Axiom 8.14); but the *choice* between the two readings remains a modeling decision, not a derived fact, and for artifacts an agent did not produce the suspension stays a convention.

(4) *L2 is witnessed, not proven.* The satisfiability of functional potential (Lemma 5.22) rests on an architectural premise – executability within an agent's operational scope – that is external to the formal system (Remark 5.23). The tag is deliberate.

(5) *No self-model layer.* The paper develops none. What Section 7.2 names – the represented symbols $\widehat{\Theta}_a$, and the conjectures Remarks 6.7 and 7.6 that no result uses – marks the distinction without developing it. On what an agent can establish about itself the paper therefore takes no position in either direction, and Corollary 6.6 is not to be read as one: it concerns a reading of K, not a self-model.

Open questions. The following are open problems. They are questions, not claims: no answer is asserted, and no epistemic tag beyond "open" is attached.

(1) *Graded generalization.* If the binary predicates are replaced by graded ones (values in $[0, 1]$ or in a lattice of partial certificates), does an analogue of Theorem 4.8 survive, and what replaces the region poset of Section 4.6?

(2) *Population structure.* For a population of agents, what is the structure of the disagreement relation induced by relative knowledge (Definition 5.9) – which distributions of region assignments are realizable, and under what witnessing conditions can every semantic collapse (Lemma 5.27 and Remark 5.28) be resolved by external witnesses?

(3) *Adjudication among witnesses.* The framework builds an external witness into functional potential (Definition 5.12 and Lemma 5.22), yet confers authority on none: the K-objective reading is itself indexed to an agent's operational scope (Definition 5.14) and its satisfying premise is external to the system (Remark 5.23). Under what conditions does one agent's certification transfer to another, and what settles a disagreement between two witnesses? Infomorphisms and information channels [BS97] are the natural machinery to connect.

11. Conclusion

Three predicates over a domain of artifacts, read per agent, record a finite datum – which of the seven regions they carve are inhabited – and this paper has asked what that datum fixes and what it leaves free. Both questions admit exact answers, and the two together are the paper's subject.

Part of the answer is classical, and it is worth marking as such. Once each region is read as the set of predicates holding on it, an agent's type is a family of non-empty subsets of $\{M, C, K\}$ whose union is the whole set, so the admissible types are the covers of the predicate set and the counts of Theorem 7.2 – 109 labelled, 34 up to relabelling – are the classical cover numbers (Remark 7.3). Neither count is claimed as a new enumerative result. What the framework supplies at that point is the identification of an agent's semantic–functional state with a cover of its own predicate set, and the interest lies in what the reading of the cells then yields over that substrate. The substrate is symmetric in the three predicates; the results below are not, because the modelling choices of Axiom 3.11 and Definition 4.16 single out one pair of them.

Over that substrate the paper proves four things. The types with empty symbolic carrier are characterized exactly, by the absence of the two overlap cells, and there are seventeen of them (Corollary 7.5); since a query's denoting token must lie in the carrier, an agent of such a type poses no query whatever architecture it carries (Corollary 8.23). The weight of that result should be stated plainly, as it is where it is proved (Remark 8.24): the carrier requirement itself is definitional, and what carries the conclusion is the composition – a finite occupancy datum, carrying no artifacts and no operations, settles an operational question uniformly over architectures, because an agent lacking a token fails the definition before any architecture-sensitive clause is reached. The coarser datum the type fixes is a six-coordinate structural profile, and its realizable values are characterized rather than enumerated: four explicit Boolean conditions decide membership in the image (Theorem 7.12), from which the census of 39 of 64 follows (Corollary 7.13) and, by the form of those same conditions, closure under coordinatewise join (Proposition 7.14 and Remark 7.15). Against these stands a boundary. The architecture layer first supplies a positive result: held fixed, internal generation reaches its fixed point at the compositional closure of its basis, after exactly the stabilization depth, and admits no further step (Theorem 8.32). The boundary is that this closure – the whole set of artifacts an architecture produces, and a summary far richer than the type – does not determine how they were produced: no function recovers from it either the generation depth of its elements or the pairs on which the composition is defined (Corollary 8.8), and the size hypothesis under which this holds is exactly the threshold at which such ambiguity can arise at all (Remark 8.9). A coarser consequence of the same construction returns to the type: two agents may carry the same complete triple, hence the same type, under different architectures, and their execution and linkage memories then differ, so neither is a function of the type (Theorem 8.30). The step relation carries no time index and no operation of supply, and so models no acquisition.

Taken together these separate what depends on what. Some objects are settled by the type alone, the structural profile among them; some require the extensions themselves, such as the contents of Θ_a and an agent's storage; execution is sensitive to the architecture; and linkage, together with the availability of a query, needs more than either. The framework does not rank these, and the distinctions are not degrees of anything: they record which data an answer requires. That is why a finite type is useful without being complete.

Two boundaries close the account. The first is one of language. Richer object languages can express an agent's knowledge that some unawareness obtains without naming what it is unaware of (Section 9.1); the query apparatus developed here has no quantified syntax of that kind, and extending it to unnamed subjects is left open rather than settled. The second is one of application. The predicates are supplied model data, and every theorem here is a statement about an arbitrary assignment meeting its hypotheses; identifying the intended assignment from external evidence lies outside the present formalism, as Section 10 records.

The finite type is therefore informative and deliberately partial. It determines exact structural constraints and settles some operational questions uniformly, while leaving the extensional and architectural degrees of freedom it discards visible as such – and knowing which questions fall on which side of that line is what a classification of this kind is for.

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